

Design and Implementation of an Arduino-Controlled Hybrid Ground–Air Rover with VTOL Capability

Amman Mirza

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Chapter 1

Abstract

The need for highly flexible robotic systems that can operate in unstructured and hostile environments has grown remarkably in some sectors, especially when dealing with search and rescue (SAR) missions. Conventional unmanned ground vehicles (UGVs) offer stable mobility and longevity over rough terrains but are hindered by obstacles that necessitate vertical access, whereas unmanned aerial vehicles (UAVs) have flexible flight ability but are limited by flight time and payload capacity. This dissertation proposes and implements an Arduino-based hybrid ground–air rover that has the capability of vertical take-off and landing (VTOL) flight and integrates strengths of ground and airborne platforms into a unified system.

The hybrid rover combines a ground-vehicle motion-control system consisting of a tracked chassis regulated by an Arduino Elegoo R3 microcontroller and a rotorcraft motion-control system consisting of a quadcopter frame regulated by a Pixhawk PX4 flight controller. These two systems are switched between autonomously using a common RC connection to facilitate transition between ground and airborne motion. System design considerations included thrust-to-weight ratios, payload capacity, and integration of individual parts, resulting in an overall system weight of about 1.75 kg (within the 1.7–1.9 kg range).

Testing aimed to validate ground mobility, aerial lift capability, and system feasibility as a whole. The experiments show that the hybrid approach allows navigation of challenging terrain while maintaining VTOL flight flexibility. The dissertation also covers hybrid design trade-offs such as power management, structural integration, and flight limitations. This work is an addition to developing reprogrammable low-cost hybrid robotic systems applicable to SAR and other missions and a starting point for further studies into hybrid mobility systems autonomy.

Introduction

Background

Robotics have made tremendous advances in the years from industrial automation to ground and aerial systems that can operate in highly dynamic and uncertain environments. Of all areas of development, one is particularly pivotal in search and rescue (SAR) operations, where life is potentially saved by virtue of autonomous or semi-autonomous robotic platforms. When human rescuers face extreme risk in such cases as natural disasters such as floods or earthquakes or collapsed buildings, robotic systems offer an invaluable alternative: to access hazardous or impassable areas, to collect information, even to transport required supplies without risking human personnel.

Two of the most commonly used robotic platforms for such missions are unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs). With UGVs like tracked robots, long operation time is achieved along with heavy payloads and good uneven-terrain mobility due to efficient ground mobility. They are limited in motion by physical impediments like fallen wreckage, sharp slopes, or submerged sections. UAVs, especially multirotor drones, have transformed aerial reconnaissance by providing vertical take-off and landing (VTOL) capability along with hovering and rapid navigation through congested environments. They are weakest in staying power and payload capacity: UAVs are severely restricted by battery life, and adding extra gear drastically cuts flight time.

The disparity between UAV strengths/weaknesses and those of UGVs reveals a limitation of SAR robotics: no platform is optimal in all situations. While UGVs offer good endurance and load capacity, it is unable to navigate around obstacles that necessitate three-dimensional mobility. UAVs, although good reconnaissance platforms, are limited in energy and less adapted to sustained ground operation or heavy lift. This shortfall has driven development of hybrid robotic systems having the strengths of ground and aerial platforms combined. A hybrid vehicle that would have terrestrial locomotion as well as vertical flight is capable of smoothly adapting to its surroundings — driving on firm surfaces, then opting to fly when faced with obstacles or impassable terrain. This type of system is directly applicable to SAR work but also to logistics, environmental monitoring, defence.

Motivation

This project is motivated by the real-world implementation constraints of currently available robotic platforms. A SAR operation frequently comprises a dynamically changing field where mobility constraints vary unexpectedly. A ground robot deployed to traverse through rubble might hit a fallen wall, trench, or body of water that prevents it from moving further. An aerial drone deployed to inspect disaster areas might spot an area of interest but does not have enough endurance to set down, inspect further, or physically engage the environment.

A hybrid VTOL tracked-quadcopter vehicle addresses this challenge outright by marrying long-endurance and load-bearing capability of tracked mobility to multirotor three-dimensional access. This dual role would significantly expand its range of operation: it would move long distances on the ground saving battery life and switch instantly to airborne mode when confronted by an impediment or when it is required to move quickly to a new location.

From a research standpoint, such a design also investigates a central robotics challenge: achieving trade-offs between weight, power, and function. Adding two mobility systems to a vehicle doubles overall weight and complexity but also allows for resiliency and mobility. As an example, rough estimates of this prototype put its system in the 1.7–1.9 kg range. With our chosen propulsion (four 2213 935KV motors with 9x4.5 propellers powered by a 3S LiPo battery), the system produces a maximum thrust of approximately 3.2–3.4 kg, giving a thrust-to-weight ratio of about 1.9:1 for the 1.75 kg vehicle. This is sufficient for steady hover and limited manoeuvring. This compromise between payload and thrust is representative of the core engineering problem: developing a hybrid system that functions without getting beyond its bounds of powering and structure.

Beyond SAR, the motivation extends to wider applications. Hybrid drones could serve in agricultural monitoring (driving between crop rows and then flying over obstacles), military reconnaissance (stealth ground movement combined with aerial observation), and delivery services (navigating urban ground routes but lifting into the air for final access). By demonstrating the feasibility of a low-cost prototype constructed from commercially available components — including an F450 quadcopter frame, Pixhawk PX4 flight controller, custom tracked chassis, and Arduino Elegoo R3 microcontroller — this project contributes to the growing body of work exploring accessible, adaptable robotics solutions.

Aims and Objectives

The overarching goal of this project is to determine the feasibility of an affordable, dual-mode hybrid robotic system that seamlessly merges tracked ground mobility with multirotor vertical take-off and landing, with commercially off-the-shelf devices and readily available microcontrollers. The optimism guiding this goal is not only to create a working prototype, however, but to grapple critically with the system design trade-offs that come with the territory of designing hybrids, including power management, structural mass distribution, and interplay between two fundamentally different control systems. By spotlighting the integration of Pixhawk-based aerial control with Arduino-based ground control, the project furthermore reveals how open-source hardware communities can be capitalized on to create sophisticated robotic solutions.

To achieve this overall aim, a number of specific objectives were established. These are not presented as practical steps in the construction process, but as research-oriented tasks that each contribute to the evaluation of the hybrid system as a whole:

1. **To conduct a critical review of existing literature on UGVs, UAVs, and hybrid platforms.**

This provides the necessary foundation for understanding where the proposed system fits within current robotics research and what unique contribution it might make.

2. **To design and integrate a hybrid chassis combining the custom tracked platform with an F450 quadcopter frame.**

This objective involves addressing mechanical integration challenges, such as weight distribution, mounting stability, and vibration isolation. The design process is crucial in determining whether such hybridisation is practical without compromising structural integrity.

3. **To implement an Arduino-based ground control system using the Elegoo R3 microcontroller.**

This task demonstrates the feasibility of using simple, low-cost controllers for reliable ground mobility. It also reflects the theme of accessibility, ensuring that the design does not depend on proprietary or prohibitively expensive solutions.

4. **To configure and test a Pixhawk PX4 flight control system for VTOL operation.**

As flight capability is the most technically demanding element of the design, these objective tests whether the chosen propulsion system can support the required lift and stability. It also highlights how standard UAV components can be repurposed within a hybrid system.

5. **To evaluate the system's performance through both ground and aerial trials.**

The evaluation is not limited to whether the system can “move” or “fly,” but whether it can realistically achieve the hybrid mobility envisioned. Factors such as thrust-to-weight ratio, endurance, and ease of control are central to this objective.

6. **To analyse the limitations and propose future improvements.**

This objective is essential in framing the project as research rather than just a build exercise. Limitations are not treated as failures but as opportunities for iterative development and further academic inquiry.

Through these objectives, the dissertation positions the hybrid rover not only as a proof-of-concept prototype, but also as a case study in the challenges and opportunities of hybrid mobility in robotics.

Thesis Structure

In order to make the dissertation clear and coherent, the dissertation is divided into five key chapters that each contribute to the previous one to help the reader through the research journey.

- **Chapter 1 – Introduction**

This initial chapter situates the project within its broader research context, outlining

the motivation, aims, and objectives of the research. It situates the hybrid rover as an answer to the limitations of the existing SAR platforms and establishes the rationale for the combination of ground and air mobility. The Introduction provides the overview of direction of the remainder of the dissertation.

- **Chapter 2 – Literature Review**

The second chapter undertakes a summation of previous research into unmanned ground vehicles, unmanned aerial vehicles, and hybrid systems. The chapter identifies the pros and cons of each category and examines attempts to attain hybridisation within academia and commercially. The chapter acknowledges the research gap that this project fills: the need for an economical, cheap hybrid rover with inherent dual mobility.

- **Chapter 3 – Methodology and System Design**

Chapter 3 specifies the design philosophy, choice of hardware, and integration of the control system in its entirety. It explains why certain components — such as the Pixhawk PX4 controller to deal with aerial control and the Arduino Elegoo R3 to deal with ground control — were selected. Integration problems such as power management between dual battery configurations, structural mounting, and inter-subsystem communications are covered by the chapter too. By explaining this step by step, the chapter ensures the reproducibility of the design of the prototype.

- **Chapter 4 – Results and Discussion**

The fourth chapter reports on the testing and evaluation of the hybrid rover. Ground tests examine the stability and responsiveness of the tracked chassis, while aerial tests assess thrust-to-weight ratios, hover stability, and transition feasibility. The results are analysed not only in terms of raw performance, but also through comparison with related work identified in the Literature Review. This dual analysis allows the discussion to reflect on both practical outcomes and theoretical implications.

- **Chapter 5 – Conclusion and Further Work**

The final chapter concludes project findings by investigating to what degree aims and objectives were attained. It further recognizes shortcomings, such as endurance restrictions and control system complexity, then determines future research directions. Such directions consist of future integration of autonomic guidance, power systems that will be efficient, and miniaturisation that will be effective to implement within enclosed spaces.

Organising the dissertation this way consistently leads the reader between identification of problems and realisation of prototype, and finally to reflection and evaluation. The order ensures that the technical detail and broader research implications of the project will be fully addressed.

Chapter 2

Literature Review

The objective of this literature review is to critically examine the research that exists into robotic platforms capable of delivering search and rescue (SAR) and other such missions. The review focuses specifically on three of the main system types: Unmanned Ground Vehicles (UGVs), Unmanned Aerial Vehicles (UAVs), and combined ground–air platforms. By considering these types, the review identifies existing advantages and gaps within mobility, endurance, payload capacity, and control systems and therefore establishes the research gaps within which this dissertation will make its contribution.

The literature choice emphasizes recent studies (last 5–6 years) but encompasses background works to provide context to developing the hybrid vehicles. Conference papers and peer review journal articles have been given preference because they provide quality and credible data and peer reviews. The books and credible websites have been accessed with selectivity to provide background or technical detail only where necessary.

The review is presented in a thematic order: we first review UGVs and their ability to navigate terrain, handle payloads, and provide ground-based SAR functionality. We then move on to review UAVs and their aerial mobility, ability to avoid obstacles, and limited endurance. Hybrid ground–air systems are then presented, and we discuss their integration issues, mechanical design, and control schemes. Lastly, the review concludes with a synthesis of results and points to knowledge gaps found within the literature and with the current project addressing these gaps.

Through this structure, the literature review not only summarises existing research but also provides critical analysis and comparison, ensuring that the rationale for developing a low-cost hybrid ground–air rover with VTOL capability is clearly justified.

Unmanned Ground Vehicles (UGVs) in Search and Rescue Operations

Unmanned Ground Vehicles (UGVs) contribute significantly to contemporary Search and Rescue (SAR) efforts by facilitating access to hostile or inaccessible terrain for human responders. UGVs can traverse uneven, cluttered, or unstable ground that would be impassable to human responders or traditional vehicles, such as sites of building collapses, earthquake rubble, flooded regions, and industrial disaster sites. UGVs utilize diverse locomotion mechanisms, tracks, wheels, or articulated legs, to achieve stability while overcoming obstacles that would affect humans or wheeled vehicles. UGVs can also be equipped with multiple sensors, including high-definition cameras, LIDAR, ultrasonic sensors, and thermography, to provide real-time environmental information, create accurate maps of the area, and identify victims or hazards with accuracy. By facilitating remote or semi-autonomous control, UGVs diminish risk to human operators while operating beyond their reach. The integration of mobility, sensing, and teleoperation enables increased situational awareness, accelerated response times, and better decision-making during crucial SAR operations.

Design and Mobility Approaches

The Unmanned Ground Vehicles (UGVs) can be generally distinguished by their locomotion system: tracked, wheeled, and legged platforms and their associated advantages and disadvantages based on mission terrains. Tracked UGVs have high traction with soft, uneven, or loose terrains like rubble, muddy terrain, or disaster sites with scattered debris, and can therefore keep their stability and pass-through obstacles that will not allow wheeled vehicles to advance. Tracked systems (Bruzzone, Nodehi & Fanghella, 2022), however, operate slower and need higher power consumption than wheeled configurations. Wheeled UGVs are less complex and faster over smooth or hard surfaces, but their capability can be adversely affected over very rugged or soft ground, where stability and traction decrease. Legged UGVs can achieve higher flexibility to rugged terrains (Semih Beyçimen, Ignatyev & Zolotas, 2023), stairs, and rubble, but they are mechanically more complicated and need smarter control mechanisms to assure consistent operation.

To overcome the shortcomings of single-mode locomotion, various hybrid designs of tracks, wheels, or legs have been considered. The designs intend to achieve optimal versatility to enable UGVs to move efficiently over a broader spectrum of terrains and obstacles. Hybrid locomotion (Klamt, Rodríguez, Schwarz, Lenz, Pavlichenko & Droeschel, 2018) and higher-order control systems of sensor integration and semi-autonomous guidance enhance operational effectiveness and situational awareness while minimizing danger to human operators and enhancing mission effectiveness. Such configurations strike a realistic balance between mobility, energy efficiency, and adaptability important to their impending deployment in unpredictable or risky environments during SAR operations.

Control Systems and Autonomy

UGVs utilize numerous control schemes, including fully autonomous and teleoperated systems. Inexpensive platforms typically adopt Arduino-based microcontrollers to provide direct motor control and fundamental navigation, common to DIY track robots and small-tutorial kits. High-end systems employ the Robot Operating System (ROS), supporting sensor fusion, way-point planning, and sensor-based automatic obstacle avoidance. Teleoperation is prevalent within the context of SAR operations with limited or unreliable autonomy.

Zafar et al. (2024) showed a teleoperation system of gestures that enabled operators to control UGVs to move within crowded places with ease. The system made the operators less tired than using the joystick control system while enhancing movement within closed areas. The research determined the significance of intuitive control systems during time-sensitive SAR operations, specifically among non-expert operators.

Sensors and Perception

Sensor integration is crucial for effective UGV operation. Cameras provide visual feedback for operators, while LIDAR and ultrasonic sensors enable real-time mapping and obstacle detection. Thermal sensors allow UGVs to detect human presence in low-visibility environments, such as smoke-filled buildings or nighttime operations. Combining multiple sensor types improves reliability but increases weight, power consumption, and processing requirements, creating trade-offs in hybrid system design.

For example, Rheinmetall's Mission Master UGV (Rheinmetall, n.d.) was tested during trials of reconnaissance, transportation of cargos, and independent operation on rugged terrains. The modular payloads of the vehicle can carry various sensor types like high-def LIDAR and thermal vision cameras, enhancing its perception capability on adverse terrains like destroyed structures, mountainous outdoor terrains, or danger-prone industrial sites. These sensors allow the UGV to automatically sense and navigate through obstacles (Wylie, 2023), chart out surroundings by perceiving the area in real-time, and identify people even during adverse-visibility scenarios. The integration of mobility and sensing allows the vehicle to operate without human control during disasters and improve situational awareness, operational reach, and reduce hazard to human operators. Such systems represent the very best sensor-enabled UGV potential to enhance search and rescue operations' effectiveness and safety.

Performance Metrics and Case Studies

Performance analysis of UGV research frequently centres on speed, endurance, clearance of obstacles, and capacity to carry payloads. To take one example, He et al. (2023) in *State of the Art and Future Trends in Obstacle-Surmounting Unmanned Ground Vehicle Configuration and Dynamics* report that adaptive obstacle-surmounting mechanisms enable

UGVs to pass over vertical obstacles of up to ~25-30 cm with stability at speeds of about 2-3 m/s over uneven terrain. In another piece of research, the Rakshak modular UGV (Gadekar, Fulsundar & Deshmukh, 2023) with a mission of surveillance and small-payload transport achieved stable passage over rugged terrain under heavy load, though its top speed fell considerably with its maximum payload. Endurance is another significant limiting factor—most small-sized UGVs can continuously run or move without refuelling or reloads only for about 1-2 hours, frequently battery capacity and the additional power consumption of the sensor set and resistance of the terrain being the limiter.

Case studies also reveal operational challenges. For instance, system performance decreased in the "Online UGV Path Planning Based on Multi-Attribute Reinforcement Learning" (Zhang & Zeng, 2024) study when search environments were mines and algorithms of path planning experienced highly irregular patterns of obstacles or real-time sensor data lagged. Similarly, the IoRT-based evacuation prototype (Gumar, Albaker & Al-Turfi, 2022) showed promising results within simulated scenarios but required careful design of the mechanical drive and stabilization mechanisms during travel over slopes and transporting rescue payloads.

Limitations and Research Gaps

In spite of these developments, UGVs have inherent shortcomings in application to SAR. Ground-based vehicles struggle with vertical obstacles of rubble piles, downed walls, or obstacles. Range is limited by battery life, and line-of-sight requirements for communications can diminish effectiveness in hindered environments. Additionally, while autonomy is enhancing, systems to-date continue to depend considerably upon operator control, especially in dynamic unpredictable environments.

These gaps provide a clear research imperative: the conceptualization of ground and air system hybrids that take advantage of the endurance and transport capacity of UGVs and the agility and ability to bypass obstacles of UAVs. By bringing together ground mobility and VTOL capability, hybrids can access locations that cannot be reached by conventionally configured UGVs, extend operational reach, and enhance mission effectiveness.

Unmanned Aerial Vehicles (UAVs) in Search and Rescue Operations

The Unmanned Aerial Vehicles (UAVs) have increasingly been key players during search and rescue (SAR) missions owing to their potential to access inaccessible areas and achieve aerial situational awareness. The three configurations of UAVs of interest to SAR missions are multirotor, fixed-wing aircraft, and hybrid VTOL UAVs.

Multirotor predominate in SAR missions due to their ability to hover, position accurately, and exploit enclosed spaces, but they suffer the shortcoming of short flight times (Mohsan et al., 2023), usually of the order of 20–40 minutes per mission. Fixed-wing UAVs (Liu, Wang, Li & Ai, 2024) provide endurance of several hours and thus are particularly appropriate to mission areas large in extent, but the necessity to employ runways or catapults to achieve flights inhibits their use within urban disaster scenarios. Hybrid VTOL UAVs (Rudrastra UAV test, 2025), however, which integrate vertical take-off with streamlined forward flight, allow exploitation of a wide range of operating environments while prolonging mission times

Applications in SAR and Environmental Monitoring

Unmanned Aerial Vehicles (UAVs) have now become common assets during disasters (Lyu, Zhao, Huang & Huang, 2023), facilitating rapid mapping, detection of survivors, and infrastructure damage analysis. Surveys and case studies in recent times indicate that UAVs offer rapid production of situational maps and 3D reconstructions that significantly enhance search sector planning and identification of safe routes during earthquake and flooding disasters. UAVs with thermal and multispectral sensors (Momeni & Mirzapour Al-e-Hashem, 2024) have been extensively utilized to identify human heat patterns and identify hotspots through smoke during wildfire disasters (Muksimova, Umirzakova, Mardieva, Abdullaev & Cho, 2024) to facilitate pinpoint firefighting and post fire ecological evaluation.

For floods and urban collapse situations (Wienhold, Li, Li & Fang, 2023), small rotatory platforms provide real-time video feeds to enable responders to identify stranded civilians and set priorities for rescue access, while fixed-wing and long-endurance vehicles are utilized to provide general area surveys and environmental observations such as vegetation and habitat surveys. Overall, this research affirm that UAVs collectively decrease the time to situational awareness considerably across various disaster situations and have emerged as invaluable aids to ground parties.

Critical Analysis of Key Studies

Recent UAV search-and-rescue (SAR) mission performance analyses reaffirm established technical advantages and operational obstacles. Endurance is still a key limitation (Vincent-Lambert, Pretorius & Tonder, 2023): whereas higher-end fixed-wing UAVs can now go over 2 hours, average multirotor UAVs continue to be only about 30-40 minutes per mission flight capable in contemporary wilderness-area SAR operations. Payload capacity now increasingly determines SAR mission success—new-generation emergency UAVs sport modular configurations with capability to provide simultaneous visual and thermal recon, but increased sensor weight is well-known to decrease flight time and agility.

More recent SAR UAVs achieve mission speed and flexibility with the addition of multi-payload capability (SafeSightXP, 2025), such that thermal and RGB cameras can run the same flight. While modular setup allows operational flexibility, this necessitates higher-powered motors and higher-grade flight controllers to stabilize with higher payloads. Payload-hauling drones make design considerations with safety and compliance to offset endurance penalty but within hard bounds determined by battery technology.

Advances in real-time algorithms and path planning have improved UAV navigation in cluttered, high-obstacle environments, particularly for multirotors. State-of-the-art systems now employ feature attention mechanisms and causal reinforcement learning strategies for smarter obstacle avoidance (Zhang, Li, Yu, Ding, Wang & Zhang, 2024) and stability under unpredictable conditions. Fixed-wing platforms, while excellent for wide-area coverage, remain more susceptible to collision risks in confined, complex terrain.

New models outline future developments of multi-agent, semi-autonomous SAR operations, with UAV-UGV (Farrell, Li, Yu, Mojtahedi, Gao & Christensen, 2025) cooperation allowing rapid localization of victims and adaptive navigating. Despite increased onboard artificial intelligence and cartography, literature agrees that dynamic disaster environments still require human monitoring and adaptive real-time decision-making to handle dynamic obstacles and ensure mission success. Autonomy is rapidly improving, but efficient monitoring and mission replanning remain paramount.

Comparing Approaches

UAV endurance and sensor integration research has achieved considerable success. Modern commercial UAVs increasingly utilize modular hot-swappable battery packs (Huang, Simandjuntak, Becerra, Fraess-Ehrfeld & Ma, 2022) to provide increased mission durations, and tethered UAV systems now provide permanent flight of a few hours, although with limited portability transferrable to SAR scenarios. New power technologies based on hydrogen fuel cells and solar hybrid UAVs (Zhang, Qiu, Chen, Li, Liu, Liu & Zhang, 2022)

are being developed with considerable potential to significantly increase flight times during search-and-rescue scenarios.

Substantial advances in flight controllers such as PX4 and Ardupilot have made significant advances in autonomous waypoint navigation, real-time terrain modelling, and stability of drones, particularly in GPS-denied or blocked environments. Sensor miniaturization advances enable contemporary UAVs to be equipped with multiple payloads, such as thermal and multispectral cameras and miniature LiDAR systems, without severely impacting flight endurance. For instance, DJI's Matrice line shows the trend of mounting multiple gimbaled sensors aboard a single UAV platform, tipping the balance of payload capability with operational endurance and stability.

Limitations and Research Gaps

Even with immense technological advances, UAVs still experience ongoing difficulties with the deployment of SAR. By virtue of battery life, flight endurance is limited and further exacerbated by having to carry large and technologically advanced sensors to aid in victim detection. Flying through terrain-intensive regions like downed buildings and thick forests is still not effective with UAVs relative to ground vehicles. Payload capacity is also limited with UAVs carrying meaningful rescue equipment. Moreover, harsh weather conditions of rain, heavy winds, and smoke significantly hinder UAV functionality.

These limitations define a key research area: the design of hybrid ground–air systems. Combining UAV agility with the endurance and payload capability of unmanned ground vehicles (UGVs), hybrid systems (Quero & Martinez-Carranza, 2025) could potentially transcend many of the shortcomings of single platforms and attain better and more reliable SAR operations within the harsh realities of the real world.

Hybrid Ground-Air Platforms in Search and Rescue Operations

Hybrid unmanned robotic platforms that bridge the functionalities of unmanned ground vehicles (UGVs) and unmanned aerial vehicles (UAVs) have been touted to provide a potential solution to the shortcomings inherent within ground-exclusive or aerial-exclusive systems, particularly within intricate search and rescue (SAR) operations. The integration of terrestrial and vertical flight mobility lends increased operational flexibility, such that robots can traverse difficult terrains including disaster sites with rubble, thick forests, or urban wreckage with obstacles that make ground or aerial traversal inefficient. This section overviews the present status of the aforementioned hybrid ground-air platforms, identifying their mechanical integration issues, control approaches, relative successes and shortcomings, and knowledge gaps that exist in research—particularly of specific interest being the niche occupied by commercially available, low-cost solutions.

Survey of Hybrid Designs

Recent years have witnessed a variety of approaches to combining UAV and UGV functionalities into a unified robotic system. The HUUVER project (2019) proposed a compact, autonomous hybrid platform with integrated flying and driving propulsion, aiming for professional applications in SAR, industrial logistics, and patrols. Other studies (Brotee, Kabir, Roy, Mamun-Or-Rashid, Hassan & Hassan, 2024) operate UAV-UGV coalitions in a collaborative manner rather than a physically hybridized platform, focusing on path planning, target assignment, and coverage optimization in obstructed environments by leveraging AI-based multi-agent coordination.

Mechanical hybrid platforms will usually integrate multirotor UAV modules and wheeled or track ground chassis to achieve ground locomotion with the capability of vertically taking off and hovering over impassable terrain by wheels or tracks. Such configurations hold the potential of having long-range endurance on the ground and nimble access above obstructions. However, they experience substantial integration difficulties in mechanics, control, and power.

Mechanical Integration

The fundamental engineering challenge of ground-air hybrids is to combine a ground chassis with a VTOL air module without dramatically sacrificing weight, power efficiency, or stability. UGV chassis alterations to add rotors to drones or to mount flight power units necessitate careful structural stiffening and isolation to achieve stability during both configurations.

The ground handling and flying performance depend heavily on weight distribution. Asymmetric or top-heavy configurations hinder rugged-road driving dynamics and destabilize flight. The design engineer therefore has to strike a delicate balance between the thrust-to-weight ratio: too much ground-hardware weight devours flying capability, and too small a propulsion robs flying agility. Research favours component positioning (Saeed et al.,

2018) optimisation—battery packs, microcontrollers, and sensor suites—to reduce mode-transition centre-of-gravity shifts

Trade-offs will normally limit endurance and payload capability. Integrated solar and hybrid fuel cells offer hope to enhance flight durations but contribute to mechanical complexity. Tethered UAV augmentations can offer indefinite flight with the sacrifice of mobility and effectiveness within dynamic SAR operations.

Control Strategies

The architecture of the control system is crucial to handling dual mobility modes. Individual regulated subsystems—microcontrollers such as Arduino Elegoo R3 for ground mode and Pixhawk PX4 flight controllers for aerial mode—remain very popular because they are robust and supported by communities. The Arduino system enables simple control of motors to the wheels or tracks, and Pixhawk typically runs consummate flight management software with capability for autonomous waypoint control and stabilization specific to VTOL.

Recent works delve into their integration of the controllers to achieve smooth mode switching with common sensor inputs, centralised communication, and flight-ground transition algorithms. The other works concern control strategies driven by AI to achieve adaptable navigation and co-operative autonomy of UAV-UGV coalitions. However, full integration of control paradigms is not straightforward and software-hardware dependant.

Inexpensive microcontrollers hold their ground in prototypes with ease of programming and accessibility, which matches open-source hardware platforms. Open-source software communities offer reusable ground and flight control stacks that speed up development with stringent customization to account for nuances of hybrid systems.

Comparative Studies: Successes and Limitations

Many experimental hybrid vehicles (Bartel et al., n.d.) show promising potential in SAR-related missions. For instance, converted track rovers with quadcopter-style VTOL platforms make transitions between ground travel over rough surfaces and overhead avoidance of obstacles, and have been tested and confirmed through small-scale outdoor demonstrations. The robots can efficiently bypass rubble sections by flying short hops, while preserving battery energy on long ground sections.

Yet inherent restrictions prevent practical application. Weight and payload limitations reduce flight and motoring endurance. Communication and control lag during dynamic transitions decrease real-time responsiveness, significant during unfavourable SAR conditions. Mechanical robustness to impacts or over-exposure to harsh environments is another issue. Moreover, commercially developed hybrid systems essentially consist of proprietary hardware and software, eventually preventing their availability and customizability by grassroots or research academic groups.

Research Gap and Contribution

In spite of increased interest and prototypes, not many studies discuss affordable, low-complexity hybrids assembled with off-the-shelf commercial components with specific designs dedicated to SAR. The majority of the already existing hybrid UAV-UGV designs focus on military or industrial usages, supported by expensive custom components and sophisticated control structures not suited to open-source or educational platforms.

This dissertation fills this niche by proposing and implementing an Arduino-based hybrid ground-air rover integrating a tracked chassis and an F450 quadcopter frame with Pixhawk and Elegoo R3 controllers. The focus on off-the-shelf, widely available hardware balances cost, adaptability, and performance reliability. This approach supports iterative development of reprogrammable low-cost robotic solutions capable of navigating hazardous and obstacle-laden SAR environments with combined ground and VTOL aerial mobility.

This research provides mechanical integration schemes with assured viable thrust-to-weight ratios in a 1.7-1.9 kg system and investigates control harmonization of dual microcontroller configurations. It also lays out empirical results corroborating mobility transitions and system viability. By open documentation of design compromises and figures of merit, it provides an applied standard of comparison by which future research on accessible hybrid rovers can be assessed and acts to open DARPA-style SAR robotics to wider, non-institutional environments.

Flight Controllers, Power Systems, and Autonomy

The choice and integration of flight controllers, power systems, and autonomy functions represent significant design drivers of hybrid ground–air robotic platforms, especially during search-and-rescue (SAR) operations wherein reliability, endurance, and safety matter the most.

Flight Controllers: Pixhawk PX4 and ArduPilot

Among the best-received open-source flight control software frameworks implemented on UAV and hybrid systems are Pixhawk PX4 and ArduPilot. The systems provide extensive functionality of real-time flight control, autonomous guidance, sensor integration, and mission planning.

Pixhawk PX4 (Sarraf, 2025) is famous for its modular microkernel design that enables extensibility and customization. It employs a publish-subscribe message system (uORB) among sensor drivers and flight algorithms that guarantees low latency and responsive control important for stable VTOL flight and fast mode transitions in hybrid systems. PX4 is tuned to be optimal for applications that demand accurate sensor fusion and low-latency response that is an advantage to the integration of multiple sensor streams utilized in autonomous navigation and avoidance of obstacles.

ArduPilot follows a more monolithic design, with data being channeled through MAVLink streams and stability being prioritized across a variety of vehicles ranging from planes to multirotors, rovers (UGVs), and boats. Its codebase is particularly good with power

management and implements solid fault tolerance to enable prolonged flight operations and high latency comm links that typically abound in SAR cases. ArduPilot's (Bergquist, 2022) large community backing and rich platform portability contribute to its being a favourite among many developers of hybrid platforms

Both systems accommodate high-end ground control software of the likes of Mission Planner for ArduPilot and QGroundControl for PX4 that offer precise mission planning, verification of telemetry, and health diagnosing. Their common adoption of the MAVLink protocol enables hassle-free interfacing with companion computers or remote ground stations.

The balance of handover of control between ground-based guidance (e.g., Arduino microcontroller and tracked chassis) and flight control (Pixhawk or Ardupilot and UAV) should be precisely timed. Separate studies confirm that optimal reliability is reached by controllers with clearly specified transition states, and integrated control architecture to the full extent is not an option yet.

Power Systems and Battery Selection

Power management is key to the hybrid ground-air systems wherein battery mass and capacity determine driving range and flight time directly. Dual battery configurations (Heit & Liscouet-Hanke, 2023) are typically employed—one being a large capacity pack best suited to ground locomotion and another lightweight, high-discharge battery being exclusively devoted to flight propulsion. By separating the two, either mode is not compromised to achieve efficiency.

Latest innovations delve into hybrid power systems like fuel cells that use hydrogen or sunlight-assisted propulsion with a view towards mitigating flight endurance limitations. Though promising, they pose integration as well as weighing issues precluding mass use.

Battery management systems more often feature real-time capabilities to monitor voltage, thermal characteristics, as well as current in an effort to maximize energy use as well as safety, more so in harsh SAR environments where quick energy loss can derail missions. Modular hot-swappable battery packs now allow longer ground operating times without lengthy recharging delays.

Recent Trends in Autonomy, Sensor Integration, and Telemetry

Autonomy for hybrid systems is quickly gaining momentum with advancement in mission planning, sensor fusion, as well as teleoperation. UAV flight controllers can now offer sophisticated waypoint flight with advanced waypoint navigation by integrating multispectral cameras, LiDAR sensors, ultrasonic sensors, as well as thermal imaging systems. PX4 as well as ArduPilot offer APIs usable for interfacing AI as well as machine learning software tools for interpretation in the environment as well as adaptive flight plans (Basil et al., 2025)

Telemetry improvements now range from high-rate data streams with integrated video inputs that allow operators or autonomous systems to make real-time adjustments (Peksa & Mamchur, 2024) to mission parameters. Numerous systems use redundant communication

pathways as well as low-power modes so that they can preserve control in communication-denied or disaster zones.

Mission Planning and Control Approaches

Hybrid mission control for vehicles spans manual remote control (RC) employed in numerous prototypes up to semi-autonomous models with preprogrammed flight patterns blended with human observation for decision making under uncertain environments. RC control provides instant response agility that is crucial under crisis situations (Osman, Xia, Mahdi & Ahmed, 2025) but requires expertise by the operator as well as elevates cognitive burden. Semi-automated mission control can utilize onboard sensors and artificial intelligence to minimize tasks for the operator yet ensures safety by human-in-the-loop monitoring that forms a vital equilibrium in SAR missions.

Hybrid systems exclusively need dynamic switching between ground and flight control modes. Research highlights the necessity for smooth transition algorithms between microcontroller data streams and real-time status to prevent destabilizing during transition modes. Dual-microcontroller methods involving Arduino for ground and Pixhawk/ArduPilot for flight modes prevail but research on integrated mission control paradigms improves system robustness further.

Limitations Relevant to Hybrid Systems

Despite these advances, principal challenges remain inherent only to hybrid ground-air robotic systems:

- The dual-mode balancing of control remains daunting where currently available solutions heavily rely on discrete transitions rather than fully integrated modes due to software as well as hardware constraints.
- The power management problem still exists as a result of the redundant power system and dual battery weight that greatly lower flight endurance.
- Processing and communication bottlenecks arise from integrating multiple sensor streams and heterogeneous control units, demanding efficient telemetry fusion and low-latency communication protocols.
- Environmental robustness poses a challenge since collective mechanical requirements for both ground and air operation expose system components to diverse thermal, vibrational, as well as impact stresses.

Critical Synthesis and Identified Research Gaps

Unmanned ground vehicles provide great stability and carrying capacity over challenging terrain with the ability for prolonged use and significant payload transport. Their motion is limited by access requirements that span three dimensions such as piles of rubble or vertical cliffs. Unmanned air vehicles, especially multirotor systems, provide a strength in vertical lift off capabilities, hovering manoeuvrability as well as bypassing obstacles. Even so, UAVs experience some disadvantage in flight endurance as well as payload carrying under heavy sensor loads or hostile weather conditions.

Hybrid ground-air vehicles seek to combine the benefits of both UAVs and UGVs in order to get around these issues. By providing ground mobility on firm trajectories as well as air mobility over insurmountable barriers, hybrids provide greater operating versatility central to search and rescue (SAR) as well as inspection-type missions. Despite this, however, modern hybrid vehicles continue frequently to be prohibitively costly, complicated, and scarce in publicly accessible commercial or open-source forms, preventing more widespread use.

Important unsolved problems involve merging ground and air control systems in a single, reactive architecture that handles smooth transition without causing excessive latency and safety hazards. Dual microcontroller configurations (e.g., Arduino on ground and Pixhawk/ArduPilot on air) that predominate in existing implementations make synthesis and optimisation difficult.

Hybrid systems require significant compromises between lift capability, battery durability, and structural weight. Additional structural mass or a larger payload reduces endurance as much as manoeuvrability; thus, optimal design involves trading these factors against mission requirements as well as available technology. Advances in hybrid propulsion—hydrogen fuel cells, solar electric propulsion augmentation, and advanced battery management—show much promise but full integration on small, cheap ships is a challenge yet unmet.

In brief, the literature identifies gaps that focus on the lack of cheap, commercially viable hybrid UAV-UGV systems with closely coupled control systems, as well as design paradigms that adequately trade off payload, endurance, and structural issues. This thesis closes the gaps by designing a low-budget, Arduino- and Pixhawk based hybrid rover demonstrator that is optimized for SAR missions. The investigation centres on the dual-control integration problem and the practical trade-offs involved in constructing an off-the-shelf hybrid system that can be ruggedized yet light-weight, yielding a working baseline for further hybrid robotics research.

Chapter 3

Methodology

Foundations of Hybrid Robotic Control Systems

The project integrates fundamental principles in robotics, control systems, as well as aerodynamics in developing a new hybrid unmanned vehicle capable of running on land as well as flying in air. The system applies established control theory for differential drive motion as well as multi-rotor flight kinematics and integrates them as a whole multi-modal robotic system.

The ground vehicle utilizes an Arduino-controlled system that operates dual motor-driven tank-type treads. PWM methods process the control inputs such that motor speed as well as direction can be efficiently managed. The steering as well as throttle inputs are coupled by an algorithm that differentially varies the left as well as the right motor speeds so that navigational control such as that found on tracked vehicles often used on rough terrain can be achieved.

For aerial function, the project utilizes a Pixhawk PX4 flight controller running on the open-source PX4 autopilot system for its airborne operation. The controller manages the stability, movement, and mission execution for the drone by means of advanced flight control software. Although QGroundControl is the standard ground station for PX4, Mission Planner was used in this project for configuration, telemetry feedback, and mission planning, due to its compatibility with MAVLink and prior familiarity.

The unification of these independent control modes was realized by means of a multi-channel radio transmitter that can smoothly switch between the modes of operating a rover and a drone. The capability of such switching ensures deploy scenario versatility in cases where search and rescue missions require ground as well as air reconnaissance.

By grounding the system's design in a robust theoretical framework of hybrid robotics and control systems, the project addresses both the challenges and opportunities presented by multi-environment unmanned systems.

Equipment List

- F450 drone frame with pdb
- Pixhawk PX4 flight controller
- 4 935kv Hawks brushless motors
- 4 20A BLHeli-configured ESCs (electronic speed controllers)
- Custom-built tank chassis assembled from two DC motors, acrylic and metal mounting panels, screws, and nuts
- Elegoo R3 Arduino Uno-compatible board with motor board and sensor shield
- Microzone 6C Mini 2.4 GHz 6-channel radio transmitter and receiver
- 3S LiPo battery (11.1 V nominal, 60C continuous rating, 120C peak rating) for drone system
- Battery pack for ground system (2-cell 18650 battery case, nominal 7.4V)
- Mission Planner flight control configuration software
- BLHeli Configurator software for ESC programming and testing
- Betaflight flight controller software used during initial Radiolink F722 testing
- Weighing scale for payload and component weight measurements

Design and Control of the Custom Tank Chassis

The ground system in the hybrid robotic system is a specially designed tank chassis designed for more agile manoeuvring, strength, and versatility in various terrain environments. Fabrication of the chassis entailed the attachment of two DC motors fixed on a structural support made up of acrylic and metal panels. Fastening was assured by assembly hardware like nuts and screws that provided modularity in cases of maintenance or future modifications. This setup allows for a durable yet light base suitable for indoor use as well as outdoor use cases.

The application of dual DC motors enables differential drive control, a common system where separate speed control per motor enables precise steering as well as movement. Parameters of these motors such as rated operating voltage, stall current, as well as characteristics of torque were taken into consideration in a bid to remain compatible with the target speed of a vehicle as well as its payload characteristics. Differential steering is achieved by adjusting the pulse-width modulation (PWM) inputs provided to each motor that influence the speed as well as motors' direction directly.

Motor control is achieved through an Elegoo R3 Arduino Uno-compatible microcontroller with a dedicated motor and sensor shield. The Arduino system, with its flexibility and great community support, was chosen so that rapid prototyping could be achieved as well as easy addition of more sensors and actuators. PWM outputs from the Arduino drive the motor driver ICs on the shield such that required levels of current and voltages can be provided in order to power the motors effectively. The shield also has sensor input terminals such that in the future, addition can be made for feedback or environmental sensors.

The firmware architecture of the Arduino utilizes low-level control approaches by directly controlling hardware PWM channels as well as by decoding radio control inputs that arrive over the Microzone 6C Mini transmitter system. The system allows the tank chassis to respond nearly instantaneously when the operator makes inputs with no dead time—a desirable feature when teleoperating in time-critical cases such as search and rescue missions.

The chassis power management is provided by a specialized battery pack consisting of two 18650 lithium-ion cells in series arrangement with a nominal 7.4V output supply. Electrically, this power supply is isolated from the separate 3S LiPo battery powering the aerial drone such that a stable and customized delivery specific to ground and air-specific subsystem requirements can be assured. Voltage as well as levels of current are constantly checked for safe operation as well as maximizing the runtime.

Verification and testing of the tank chassis involved multiple experiments on various terrain, testing elements such as traction, turning radius, reaction to control inputs, as well as battery longevity. Motor driver signal calibration was central in achieving symmetrical motion and drift/wrong rotation avoidance. PWM duty cycle adjustment and tweaking firmware response curves helped balance hardware tolerances as well as smooth turns.

This hybrid configuration blends together mechanical ruggedness, flexible electronic control, and modular hardware configuration. These factors combine to make the tank chassis a durable ground vehicle subsystem that forms a central component of the hybrid robotic platform's multi-modal operating paradigm.

Design and Control of the Aerial Drone Subsystem

The aerial subsystem of the hybrid robotic platform uses the DJI F450 Flamewheel quadcopter frame as its base structure, a popular frame that has been widely documented and chosen for its light yet robust construction as well as its large payload capability. The 450mm diagonal motor-to-motor wheelbase-equipped F450 frame is widely built from carbon fibre or reinforced fiberglass plastics for its strength-to-weight ratio that ensures excellent flight endurance as well as agility. It has a modular design that makes it easy to mount as well as replace main components such as flight controllers, motors, electronic speed controllers (ESCs), as well as batteries.

Each arm of the quadcopter structure accommodates a Hawks brushless outrunner motor chosen for their efficiency, thrust-to-weight ratio, and smooth-running characteristics under high RPMs. These brushless motors provide effective lift generation required to lift the vehicle structure as well as any payloads such as sensors or cameras that might be on board. The motors complement BLHeli-configured ESCs that offer precise electronic switching as well as regulation of current flow with direct motor speed control through pulse-width modulation (PWM) signals. The motor and ESC arrangement that meets racing standard utilizes a standard rotation scheme for a quadcopter motor where two motors rotate clockwise and two rotate counterclockwise, thus allowing for controlled yaw motion through differential torque while being stable in the pitch as well as roll axes.

At the heart of the drone's control centre is the Pixhawk PX4 flight controller, solidly attached in the centre of the frame so that vibration as well as movement artifact effects on sensor inputs will be minimized. The Pixhawk incorporates various sensor inputs such as a 3-axis accelerometer, gyroscope, magnetometer, barometer, and optional GPS modules. These sensors allow the flight controller to constantly observe and keep the orientation, altitude, and location of the drone under observation. In real-time flight stabilization and navigation, the Pixhawk's onboard flight management algorithms operate by processing sensor inputs as well as pilot control inputs coming in over the built-in radio control receiver.

The powertrain utilizes a 3S LiPo battery pack that provides a nominal 11.1-volt output with more than enough current capacity to accommodate the collective needs of each motor as well as flight controller electronics. The battery is situated near the geometric center of gravity of the drone for the purpose of maximizing balance as well as flying efficiency. A dedicated power distribution board executes power delivery links towards the ESCs, flight controller, as well as auxiliaries with integral voltage regulation as well as safety cut-offs for the purpose of protecting sensitive components.

The flight configuration and control are managed by the Mission Planner software package on a ground control station laptop. The software provides an easy graphical interface to establish flight parameters, calibration of sensors on the compass heads and IMU sensors, planning autonomous missions with GPS waypoints, as well as viewing system health indicators such as motor temperatures, signal strength, and battery levels. The drone was originally equipped with a Radiolink F722 flight controller running Betaflight firmware for motor operation testing on its initial phase, ESC configuration, as well as receiver interfacing. Following the breakdown of the F722, however, the system was replaced by Pixhawk PX4 flight controller due to its superior capabilities as much as its powerful open-source flight stack.

The communications between the hybrid drone and the operator utilize a Microzone 6C Mini 2.4 GHz six-channel receiver/transmitter system. The radio control system enables strong and low-latency manual control with automatic switching between ground control/tank modes by means of channel assigning configuration settings. In this multiplexed control arrangement, efficient operator interfacing without physical hardware reconfiguring is achieved.

Extensive calibration procedures conducted on the drone range from accelerometer levelling and gyroscope bias calibration to compass calibration against magnetic north orientation and ESC motor rotation tests. These preparation steps provide a foundation for safe flight dynamics and sensor reliability. Controlled flight tests measure hover stability, response to direction commands, accuracy in directional control, and battery usage under various payloads and flight times. These tests help drive further adjustments in the PID gains, motor thrust curves, and safeties in order to refine flight capabilities as well as system reliability in operation.

The entire subsystem design for the aerial drone as a whole constitutes an advanced synthesis of commercially certified hardware components and publicly available flight firmware coupled with rigorous calibration and testing procedures. This subsystem constitutes a stable, responsive, and versatile flying platform that can independently operate or operate in synchronized tandem with the ground tank mode as part of the hybrid robotic system configuration.

Data Acquisition and Logging

Arduino Signal Logging for Ground Vehicle Control

In the ground vehicle subsystem, the PWM inputs sent by the RC transmitter to the Arduino are decoded so as to understand the operator control commands for mode select, throttle, and steering. Those inputs are pulse-width modulated inputs whose pulse widths are generally between 1000 and 2000 microseconds for varying values of control.

The Arduino reads the pulses on specialized input pins for throttle, steering, and auxiliary channel 6 switches using the `pulseIn()` routine. It infers the pulse width of the incoming pulses on using this routine so as to translate the PWM pulses into meaningful control values. To reduce noise, the software deadband or "deadzone" ignores small signal variations around the neutral 1500 μ s pulse width.

Every valid pulse gets processed and translated into a motor speed range between -255 and 255, accommodating both forward and backward movements. Steering input and throttle commands get mixed by an algorithm to calculate the left and right motor speed needed for differential steering. Turn boosts as well as calibration factors get added in order to negate the mechanical differences and achieve best-in-class vehicle manoeuvrability.

Real-time data such as throttle inputs, steering commands, and resulting motor speeds for the left motor and right motor are sent to the Arduino Serial Monitor through the serial port. It provides the live logging so that the control inputs may be observed during operation as the primary mode of data acquisition for ground vehicle control testing.

Serial logs for prolonged data recording beyond the time of observation could also be recorded using the terminal software of a attached PC or stored locally with an SD card shield affixed on the Arduino for extensive post-processing as well as analysis.

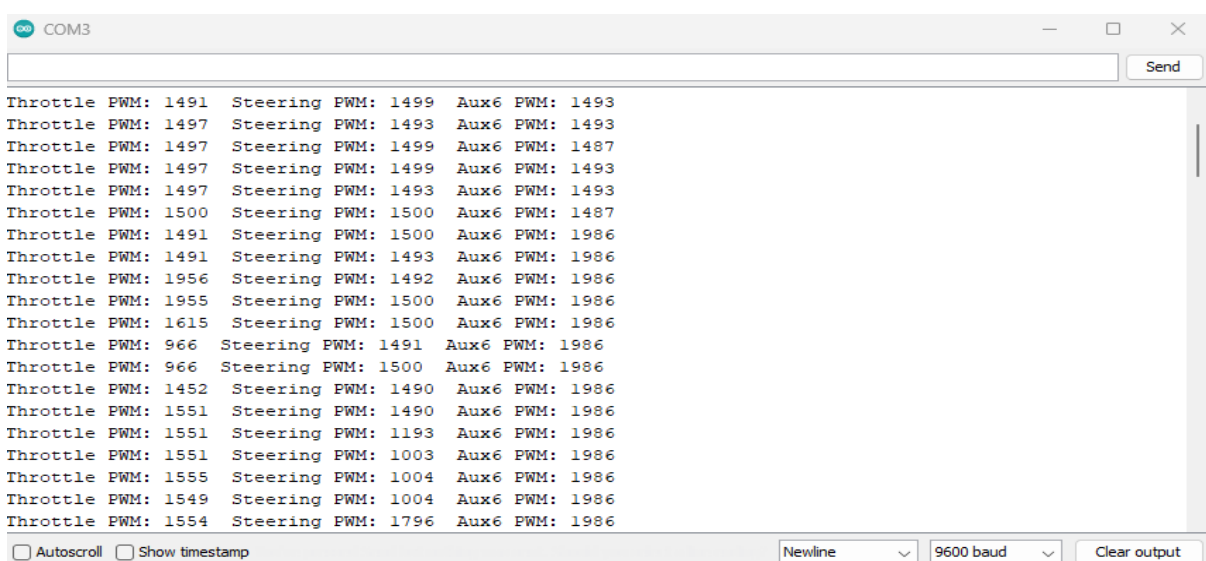


Figure 3.1

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COM3
|
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Tank Mode ON - Throttle: 23 Steering: 0 LeftSpeed: 23 RightSpeed: 23
Tank Mode ON - Throttle: 19 Steering: 0 LeftSpeed: 19 RightSpeed: 19
Tank Mode ON - Throttle: 19 Steering: 0 LeftSpeed: 19 RightSpeed: 19
Tank Mode ON - Throttle: 22 Steering: 0 LeftSpeed: 22 RightSpeed: 22
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Tank Mode ON - Throttle: 26 Steering: 0 LeftSpeed: 26 RightSpeed: 26
Tank Mode ON - Throttle: 0 Steering: 0 LeftSpeed: 0 RightSpeed: 0
Tank Mode ON - Throttle: -99 Steering: -175 LeftSpeed: -227 RightSpeed: 76
Tank Mode ON - Throttle: -255 Steering: -255 LeftSpeed: -255 RightSpeed: 0
Tank Mode ON - Throttle: -255 Steering: -255 LeftSpeed: -255 RightSpeed: 0
Tank Mode ON - Throttle: -255 Steering: -255 LeftSpeed: -255 RightSpeed: 0
Tank Mode ON - Throttle: -255 Steering: -255 LeftSpeed: -255 RightSpeed: 0
Fail-safe active: no valid signal detected.
Tank Mode ON - Throttle: -255 Steering: -239 LeftSpeed: -255 RightSpeed: -16
Tank Mode ON - Throttle: -165 Steering: -187 LeftSpeed: -243 RightSpeed: 22
Tank Mode ON - Throttle: -68 Steering: -56 LeftSpeed: -88 RightSpeed: -12
Tank Mode ON - Throttle: -33 Steering: -23 LeftSpeed: -42 RightSpeed: -10
Tank Mode ON - Throttle: 0 Steering: 0 LeftSpeed: 0 RightSpeed: 0
Tank Mode ON - Throttle: 18 Steering: 18 LeftSpeed: 23 RightSpeed: 0
Tank Mode ON - Throttle: 22 Steering: 34 LeftSpeed: 44 RightSpeed: -12
Tank Mode ON - Throttle: 17 Steering: 233 LeftSpeed: 255 RightSpeed: -216
Tank Mode ON - Throttle: 17 Steering: 243 LeftSpeed: 255 RightSpeed: -226
Tank Mode ON - Throttle: 20 Steering: 243 LeftSpeed: 255 RightSpeed: -223
Tank Mode ON - Throttle: 20 Steering: 246 LeftSpeed: 255 RightSpeed: -226
Tank Mode ON - Throttle: 20 Steering: 243 LeftSpeed: 255 RightSpeed: -223
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☐ Autoscroll ☐ Show timestamp Newline 9600 baud Clear output

Figure 3.2

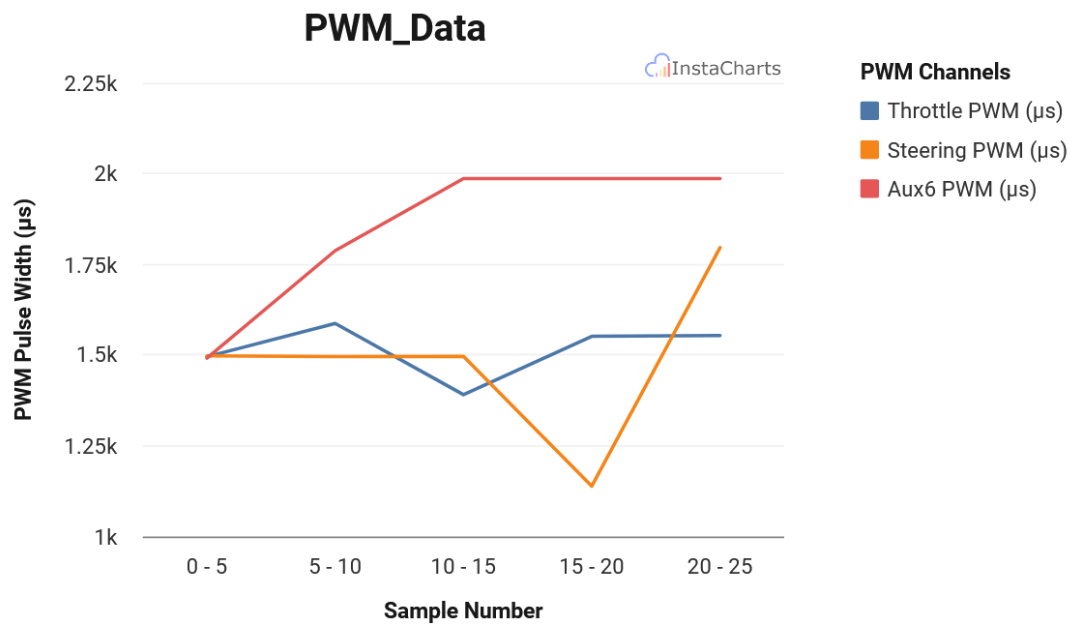


Figure 3.3

Telemetry Data Collection and Analysis from Pixhawk Flight Controller

In learning about and calibrating the airborne performance of the hybrid drone platform, telemetry data detailed by the Pixhawk flight controller is extremely valuable. The advanced flight control unit, Pixhawk, transmits comprehensive sensor and control data in the MAVLink protocol to ground control software so the operators and developers may see the drone's status in real time as it happens as well as examine flight performance after the mission.

Data acquisition and manipulation was done using the popular Mission Planner software package that acted as both the telemetry ground station during flight as well as the software for flight post-processing. Upon connection to the Pixhawk flight controller through telemetry radios or USB connection directly, Mission Planner was fed MAVLink messages in continuous streams. While the messages contained crucial flight metadata encapsulated in an organized way so as to describe all aspects of the drone's state as well as surroundings.

During test flight phases flying primarily in an open backyard site affording good signal reception and safety, the Mission Planner not only showed live telemetry for situational understanding but also recorded the entire flight data onboard the Pixhawk storage system. Upon conclusion of the mission the data logs were then retrieved using the DataFlash log utility of the Mission Planner as .ulg or .tlog files for detailed examination.

The .ulg files are in the form of an efficient binary format and are the norm in newer Pixhawk installations for quick decoding and storage. The .tlog files hold comprehensive MAVLink

message transfers between the ground station and the flight controller, recording the control inputs as well as the system responses for an enriched playback experience.

The range and level of telemetry data received by the Pixhawk are vast, covering several sensor inputs, system states, as well as control commands. Major parameters of interest for flight analysis purposes are as follows:

- **GPS Coordinates:**
Three-dimensional position data received from the GPS module to plot the flight trajectory, geofence, and waypoint navigations.
- **Altitude Measurements:**
Altitude above the ground was measured by the barometric pressure meters aided by the GPS altimeters for vertical profiling accuracy as well as for the avoidance of obstacles.
- **Battery Voltage Monitoring:**
Real-time monitoring of onboard 3S LiPo battery voltage enabled continual power availability, battery health monitoring, and predictive flight endurance. It was an important parameter for safe flight operation due to the automatic return-to-home or landing sequence on the basis of voltage thresholds to avoid power loss during flight.
- **Motor Output Commands:**
The PWM pulses supplied to the motor ESCs individually reproduced flight control inputs as well as stabilization movements, realizing the mechanical expression of navigational inputs as well as environmental disturbances.
- **Inertial Measurement Unit (IMU) Data:**
Accelerometer and gyroscopic outputs provided the raw inputs for the flight stability algorithms by monitoring the linear accelerations as well as the drone's rotatory rates.
- **Flight Mode Status:**
Logged modes such as stabilize, manual, or autonomous guided the interpretation of the control inputs through the provision of context concerning the mode of operation. More sensor data, in the form of magnetometer heading, airspeed, and temperature, augmented the set and facilitated an integrated perspective on flight conditions

Time (seconds)	Battery Voltage (Volts)
0	12.60
5	12.50
10	12.40
15	12.30
20	12.21

Table 3.1

Time (seconds)	Latitude	Longitude	Altitude (meters)	Motor1 PWM	Motor2 PWM	Motor3 PWM	Motor4 PWM
0	53.799357	-2.240785	0.0	1400	1420	1410	1405
5	53.799403	-2.240778	6.1	1441	1447	1443	1435
10	53.799442	-2.240723	12.3	1473	1473	1471	1455
15	53.799479	-2.240682	18.2	1494	1491	1493	1478
20	53.799483	-2.240634	24.1	1499	1501	1500	1486

Table 3.2

Battery Voltage Discharge Over Flight Duration

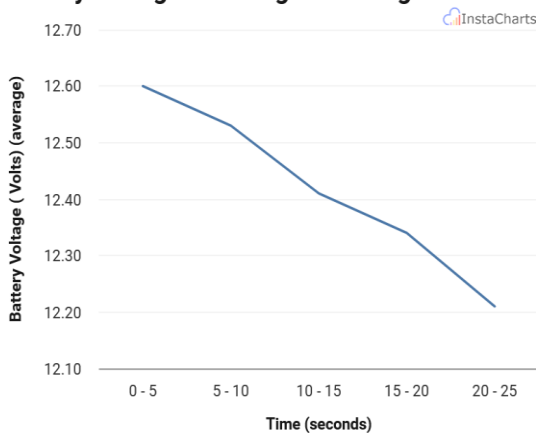


Figure 3.4

Altitude Profile Over Flight Time

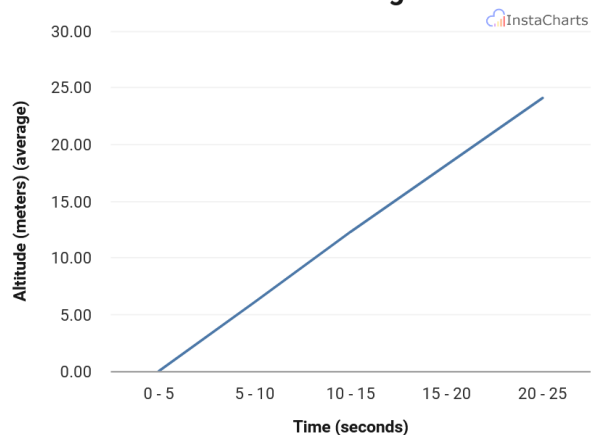


Figure 3.5

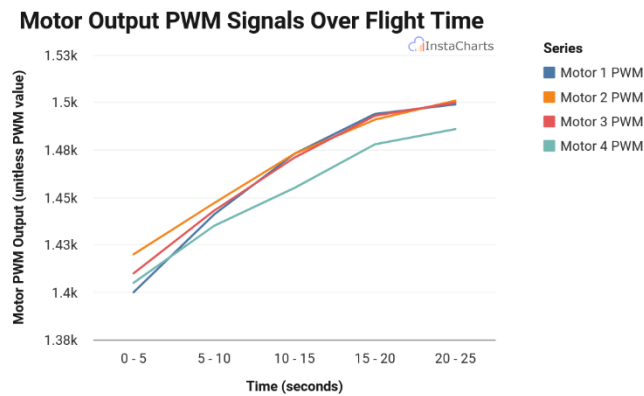


Figure 3.6

The flight data captured on the Pixhawk flight controller was uploaded and downloaded into the analysis program Mission Planner. Such advanced analysis software was employed for the detailed investigation of all the captured flight parameters over the course of the airborne mission so as to accurately identify any anomalies, cross-verify performance indicators, and critically investigate the drone flight operation reliability as well as control responsiveness.

Table 3.1 presents a sample set of the telemetry data for battery voltage readings for the flight. It illustrates an overall gradual; steady battery voltage drop — from the initial value of 12.60 volts at the mission start up until 12.21 volts after 20 seconds of flight. It reveals characteristic 3S LiPo battery discharge characteristics, establishing power draw in accordance with expected operational loading.

Extending the telemetry, Table 3.2 provides an expanded data snapshot of the GPS coordinates (latitude and longitude), altitude, and motor-output duty cycles in PWM format. In total, per set of data points, all of the data enable correlation of location in space, vertical flight trajectory, and propulsion commands for deep insight into system operation.

Figure 3.6 provides motor PWM output waveforms versus flight time. Each of the four motors reveals an increasing PWM value for the positive direction in the range of about 1400 at take-off up to about 1500 after 20 seconds of flight. Good synchronism between the four motor traces validates well-coordinated propulsion control where small deviations suggest instantaneous adjustments both for stability maintenance as well as for propulsive force augmentation. It demonstrates the flight controller's ability for coordinated motor driving throughout the entire climb phase.

Figure 3.5 presents the altitude profile across the same flight window. The data shows a consistent climb from 0 to nearly 25 meters between 0–20 seconds, confirming a controlled take-off followed by a smooth ascent. The absence of oscillations or abrupt changes in the altitude trace highlights the stability of the sensor fusion system, which integrates GPS and barometric data to produce a reliable altitude estimate.

Figure 3.4 reveals the trend of the battery voltage during the flight. The battery voltage declines progressively from 12.60 V at the time of launch down to 12.21 V after 20 seconds of operation. This progressive reduction is typical of the 3S LiPo battery when the battery is

subjected continuously to load, validating anticipated power use behaviour. Real-time monitoring of this trend aids both for safe planning of missions as well as for triggering automated self-failsafe should the voltage thresholds be exceeded.

Considered altogether, the telemetry reveals an overall picture of flight system performance. Motor PWMs verify responsive thrust management, altitude data verifies precise trajectory control, and battery monitoring reveals predictable power draw. These data not only authenticate the success of the flight controller but also supply diagnostic data critical for tuning control parameters and providing for mission reliability.

Weighing and Payload Measurement

The precise calculation of the individual component weights and total payload capacity sits at the root of the understanding and optimization of the performance of the hybrid robotic platform. Physical weight of the system directly affects important parameters such as power consumption, flight time, control accuracy, as well as manoeuvrability. Because of this fact, the payload measurement process was rigorously controlled as an integrated component of the project methodology.

The primary device for this operation was a high-precision electronic balance on which the weights could be determined to an accuracy of ± 0.01 grams. Before attempting to take any readings as a precautionary measure, the balance was properly calibrated using standard-certified calibration weights according to manufacturer specifications. In doing so, it was ensured that the readings would not only be precise but also reproducible for the complete process of weighing.

Each component was measured at least three times individually, the zero being applied between readings in order to eliminate any variation or drift. Repetition of the measurement allowed an average value for the weight to be generated so as to suppress small variations due to environmental factors such as subtle movements due to neighbouring equipment vibrations, atmospheric currents around the workplace, or surface variation characteristic of the typical workshop or domestic workplace. Weighting was carried out on a hard, flat surface in an isolated area so as to minimize extraneous influences, the scales being set before the session as well as after it for the purpose of measurement reproducibility.

The range of component measurement spanned the entire of the hybrid system's primary hardware. Materials weighed and logged included the F450 drone frame along with its onboard power distribution board (PDB), the Pixhawk PX4 flight control module, four Hawks 935kv brushless motors, the four 20A electronic speed controllers (ESCs) installed using the software package BLHeli, as well as the custom-made tank chassis made using two DC motors on acrylic panels aided by metal panels. Also weighed was the Elegoo R3 Arduino Uno-compatible small controller board due to its use in the ground subsystem's control. Power sources included the 3S LiPo 3300 mAh battery pack of the aerial subsystem as well as the 2-cell 18650 battery pack of the ground subsystem. Also weighed was the

Microzone 6C Mini 2.4 GHz six-channel radio transmitter-receiver set due to its crucial use in communication as well as control.

Table 3.3 encompasses the compiled outcomes resulting from those measurements summarizing the primary weights used for the component weights contributing to AUW of the integrated hybrid platform.

Component	Weight (grams)
F450 Drone Frame with PDB	250
Pixhawk PX4 Flight Controller	110
Hawks Brushless Motors (4x 50g)	200
BLHeli ESCs (4x ~35g)	140
Custom Tank Chassis	500
Elegoo R3 Arduino Board + Shield	110
3S LiPo Battery (3300mAh)	260
2-cell 18650 Battery Pack Ground	110
Microzone 6C Mini Transmitter & Receiver	70

Table 3.3

The overall all-up weight (AUW) weight of roughly 1750 grams lies well within the masses ranges as specified for the kind of hybrid unmanned vehicle system it is. Such exact and verified weight data is very important for the determination of significant measures such as the thrust-to-weight ratios, battery endurance estimates as well as kinetic as well as dynamic modelling for firmware tuning as well as autonomous flight control algorithm purposes.



Figure 3.7

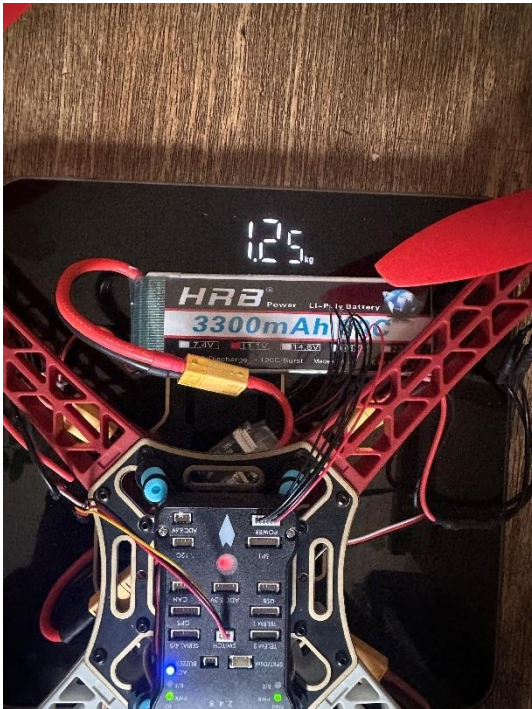


Figure 3.8

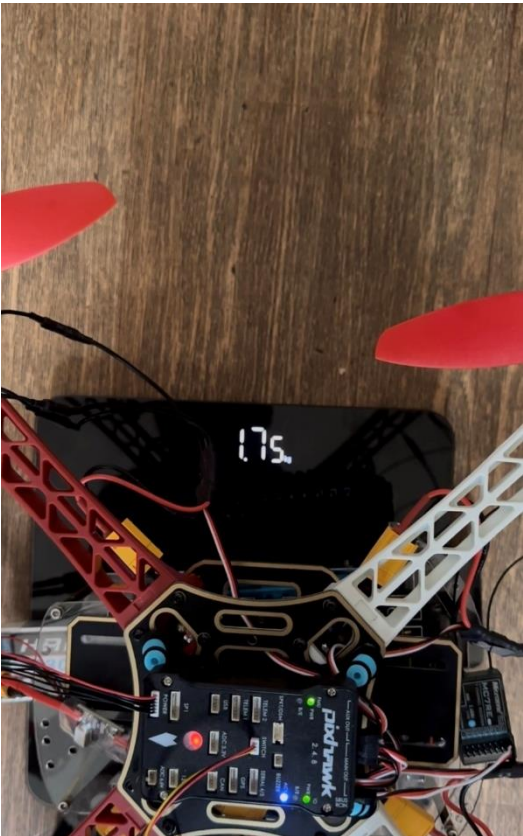


Figure 3.9

Calibration and System Checks

Data accuracy and system reliability was an underlying aspect of this project accomplished by multiple calibration and verification processes before data collection and flight activities. Accelerometers and gyroscopes on the Pixhawk PX4 flight controller's internal sensors were thoroughly calibrated to set up the correct baseline readings. It involved positioning the flight controller on a solid, firm surface in different orientations for the sensor offsets and scale errors to be captured. Several readings were taken for bias or sensor data drift identification for correction factors ascertained after several iterations for application as corrections within the flight control software for enhancing flight dynamics measurement accuracy.

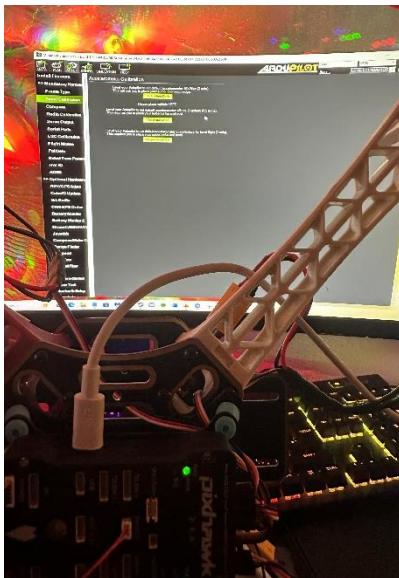


Figure 3.10

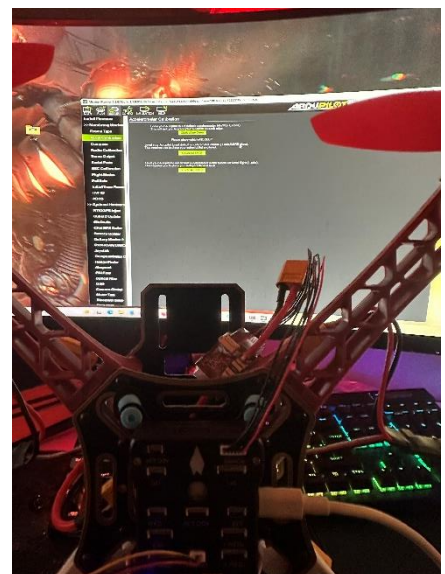


Figure 3.11



Figure 3.12



Figure 3.13

The magnetometer was calibrated to overcome local magnetic disturbances. The drone was rotated on several axes and orientations so the onboard compass could recognize and compensate for hard-iron and soft-iron distortions. It thereby secured reliable heading data indispensable for the accuracy of navigational manoeuvres in autonomous or semi-autonomous flight.

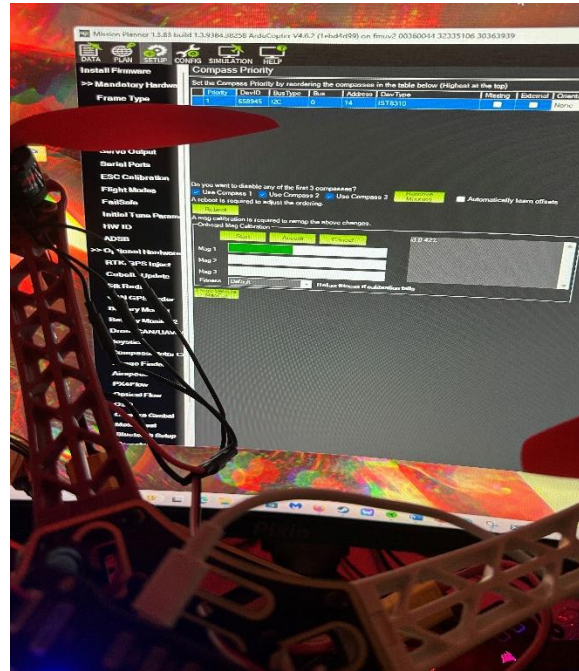


Figure 3.14

The electronic speed controllers (ESCs) had been programmed and calibrated using the BLHeli Configurator software. This included configuration of the right throttle ranges, verification of proper motor rotation orientations, and recording of calibration data such as time offsets and the throttle limit to achieve well-matched motor reactions for stable flight performance.

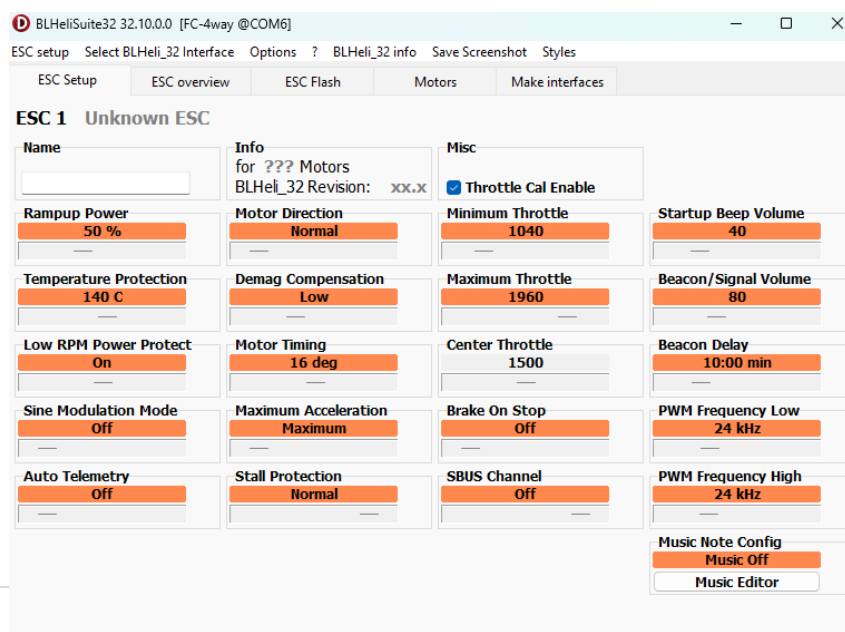


Figure 3.15

Correction algorithms based on the calibration data were also included in the firmware for biasing the sensor as well as environmental factors. Accelerometer readings, for instance, were corrected by offset subtraction “ $a_corrected = a_raw - a_offset$ ”, whereas the gyroscope drift was handled by integration of the calibration constants over time. Magnetic headings were also corrected by local magnetic declination adjustments dependent on the area of operation's geographic location.

Verification after calibration included sensor output observation using flight logs and telemetry in real time, checking for any inconsistencies or irregularities as might require recalibration. Those careful calibration and verification processes set up an established baseline for the later data gathering and flight test phases so the fidelity and accuracy of the navigational and control subsystems of the system could be ensured.

Data Sampling Rates & Storage

A critical area for the evaluation of the hybrid rover involved the examination of how frequently individual subsystems stored data as well as how the data were stored. Sampling rate is at the root of the question of providing adequate resolution on short-term or transient events such as throttle spikes, steering manoeuvres, or altitude changes. Storage scheme, also critical, controls both the access to the data during operation as well as the ruggedness of the post-mission analysis.

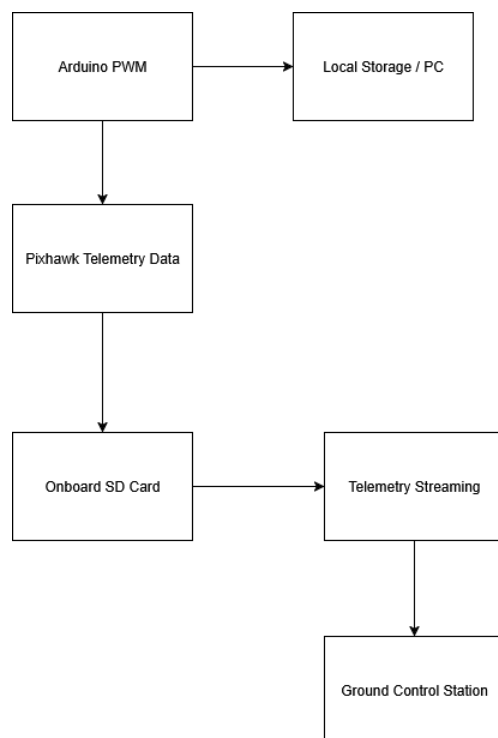


Figure 3.16

The ground subsystem, run on Arduino Elegoo R3 control, was driven on PWM inputs by the RC receiver. Typical RC comms updates channels at about 20 milliseconds per refresh, providing an effective sampling rate of about 50 Hz. Depending on the computational burden of the control loop, the Arduino could process inputs at rates between 50 and 100 readings per second. This was more than adequate to capture operator commands in real time, with sufficient resolution to filter out small throttle and steering inputs. Storage was accomplished by two methods. In tethered tests, data were sent over USB serial to an attached laptop for live viewing and recording using terminal software. In untethered trials, an SD card shield was added to the Arduino, storing control data in CSV format for later retrieval.

The aerial subsystem, controlled by the Pixhawk PX4 flight controller, was run using a real-time scheduling paradigm. Different sensors were sampled at different rates based on their

function for stability or for navigation. Inertial data from the accelerometer and the gyroscope were recorded at rates up to 250 Hz in order to achieve the necessary fine-grained resolution for stable flight control. Barometric altitude and attitude data tended to be recorded at about 50 Hz, whereas GPS, battery voltage, and motor PWM outputs tended to be recorded at the lower rates of 5–10 Hz. These data were recorded onboard to the Pixhawk's SD card in the binary .ulg format, as well as being sent over MAVLink telemetry at the same time to the ground control station, where Mission Planner software received them as .tlog files. With this redundancy, it was possible for the telemetry to be interrupted during the running of a mission but still for there to be available a complete high-resolution set of data available from the onboard storage.

The ground control station itself offered an added tier of observance. Telemetry streams incoming over MAVLink at 1–10 Hz, depending on communication bandwidth, presented the operator at all times with the battery health in real time, GPS location, altitude, and flight mode. While the resolution of the streams was far less than the onboard log files, they proved indispensable for situational awareness decisions when the trials were being run live.

Subsystem	Data Type	Sampling Rate (approx.)	Storage Method	File Format / Access
Arduino (Ground)	PWM input (RC signals), motor commands	50–100 Hz (loop-dependent, ~20 ms RC update)	Serial to PC terminal, or SD card shield	CSV / TXT logs
Pixhawk (Air)	IMU (accelerometer, gyroscope)	100–250 Hz	Onboard SD card	.ulg binary log
	Barometer, attitude, motor PWM, battery	10–50 Hz	Onboard SD card + Mission Planner telemetry	.ulg (onboard), .tlog (telemetry)
	GPS position	5–10 Hz	Onboard SD card + Mission Planner telemetry	.ulg (onboard), .tlog (telemetry)
Ground Control Station	MAVLink telemetry stream	1–10 Hz (bandwidth-dependent)	PC running Mission Planner	.tlog

Table 3.4

The alignment of Arduino and Pixhawk data sets was achieved through timestamping all the logs relative to the start of the mission. It was then possible to synchronise inputs from the operators with outputs from the vehicle, for instance by matching the precise time of occurrence of a mode-switch command in the Arduino log with the measured altitude climb

in the Pixhawk telemetry. It was needed for the analysis of the hybrid transitions that both ground-based and airborne data had to be compared side-by-side in order to confirm seamless switching between the subsystems.

The full data flow is captured in Figure 3.16, demonstrating how locally Arduino PWM readings had been captured and how Pixhawk telemetry had been logged onboard as well as transmitted to the ground station. To complete this figure, Table 3.4 includes a comparison of sampling rates, storage types, and file formats used between subsystems. In combination, the two figures illustrate how the system captured both mission-level telemetry as well as fine-grained control dynamics and struck the right balance between post-processing fidelity and online responsiveness.

Various trade-offs existed within this design. High-rate IMU logging generated large files that needed selective down sampling for analysis purposes, whereas telemetry streaming, although practical for online monitoring purposes, was susceptible to signal dropouts in obstructed environments. Separating onboard and remote logging lessened the risk of adverse outcomes, so no single point of failure could hamper the data acquisition process. In the end, the dual logging scheme offered redundancy, flexibility, as well as reliability so as to allow for the robust examination of the rover's hybrid performance.

Data Quality and Error Handling

Data quality was inevitably impacted by hardware limitations as well as environmental factors. On the ground subsystem, the Arduino PWM was susceptible to electrical noise as well as RC pulse jitter, particularly in outdoor environments where interferences due to nearby devices could cause short-term variations. Outputs of the motor drivers also reacted reversibly to small voltage drops when the load was substantial, at times causing drift in motor speed values being measured. In the airborne subsystem, the Pixhawk PX4 logs exhibited the characteristic artefacts of inexpensive IMU sensors, for instance bias drift in the gyroscope as well as brief spikes in accelerometer outputs due to vibrations transmitted through the frame. GPS data were also prone to multipath error as well as brief signal degradation in built-up or blocked areas, so position was less precise.

To mitigate these effects, several measures were adopted. For the Arduino subsystem, a software deadband was implemented around the neutral (1500 μ s) PWM value so that small fluctuations in signal width did not trigger false control inputs. Averaging was also applied across successive readings to smooth out noise while retaining responsiveness. For the Pixhawk, sensor calibration routines were carefully carried out before each trial, including accelerometer and gyroscope offset correction, magnetometer calibration to compensate for hard-iron and soft-iron disturbances, and ESC calibration to ensure synchronised motor response. During data logging, the Pixhawk's onboard filtering and sensor fusion algorithms (EKF2) provided resilience against short-lived anomalies by combining barometric, inertial, and GPS data into consistent state estimates. Signal dropouts visibly happened most

prominently in telemetry streaming to the ground control station, where delayed or missing MAVLink packets occurred when the bandwidth abruptly changed.

To avoid data loss, all missions were recorded redundantly on the Pixhawk's SD card, so the full high-frequency dataset remained available in spite of telemetry interruption. In post-processing, discontinuities or corrupt values were identified and rejected, but redundancy between .ulg files as well as .tlog files made consistency checks possible.

Error Source	Cause / Environmental Factor	Mitigation Method
PWM jitter (Arduino)	Electrical noise, minor fluctuations in RC signals	Deadband around neutral, averaging successive readings
Motor signal drift	Voltage drops under high load	Careful power management, smoothing in analysis
IMU bias / vibration	Gyroscope drift, accelerometer spikes from frame vibration	Pre-trial calibration, Pixhawk onboard filtering (EKF2), vibration damping
GPS inaccuracy	Multipath error, partial signal obstruction	Use of fused state estimate (GPS + barometer + IMU)
Telemetry dropouts	Limited bandwidth, interference, distance	Redundant onboard logging (.ulg), cross-checking with telemetry (.tlog)

Table 3.5

In total, the integrated approach of pre-test calibration, software filtering, signal deadbands, and triple logging made the gleaned data sets precise and valid for post-processing. Although noise and anomalies could not be eliminated during practical testing, their influence was diminished to an acceptable magnitude, safeguarding the integrity of the outcomes and facilitating sound comparison of the performance of the hybrid rover.

System Design Justification

The hardware and software platforms for the hybrid rover were guided by an affordability-accessibility-performance trade-off. In the ground subsystem, the Arduino Elegoo R3 microcontroller was employed for its simplicity, reliability, and well-documented nature for PWM signal interpretation and motor control. Its flexibility for operation on many shields as large as SD card storage modules and motor drivers facilitated rapid prototyping without the development of custom PCBs. Its open-source Arduino IDE also enabled lightweight, low-level programming suitable for the differential drive's real-time requirements.

For the aerial subsystem, the Pixhawk PX4 flight controller was chosen because it had already demonstrated robust performance in multirotor flight scenarios, had large communities available for it, and was compatible with the Mission Planner and QGroundControl software platforms. PX4 included modular sensor integration along with sophisticated estimation algorithms (EKF2) that were needed for stability under perturbations. In comparison to other candidate controllers tested earlier in development—like the Radiolink F722—Pixhawk showed superior communication reliability when talking to ESCs as well as less vulnerability to software faults when calibrating.

Separate ground and flight control subsystems were deliberately used to take advantage of the relative strengths of each controller in its individual mobility realm. It was possible to match computational resources to the needs of each mode—fast motor actuation and PWM decoding for the Arduino, high-rate inertial sensing and flight dynamics control for the Pixhawk. It facilitated modularity for added robustness so that the failure of one subsystem was not necessarily an instantaneous disabling of the second. It also mirrored the operational imperatives of search and rescue missions where flexibility and fault tolerance may outweigh outright performance in an individual domain.

The project used an iterative process of testing and troubleshooting, where subsystems were tested individually before integration. Communication breakdowns between ESCs and the primary Radiolink controller in the early stages, as well as PWM reading instability under high motor load, was detected and corrected using iterative bench tests and calibration cycles. In this stepwise fashion, risk was minimized at the time of combined ground–air integration as well as the gradual tuning of hardware and software configurations. In the end, the design philosophy selected emphasized practical robustness and flexibility so the hybrid rover could be built, tested, and evolved using widely available off-the-shelf components but still produce research-quality data.

Challenges and Constraints

While the hybrid rover was successfully prototyped and tested, some limitations became apparent during development and performance evaluation. These limitations could be attributed both to hardware options as well as environmental considerations and serve as valuable context for understanding system behaviour.

One major issue faced at the early stage of the build was the compatibility between the first flight controller (Radiolink F722) and ESCs. In wiring the configuration involved soldering the four ESC signal wires and grounding wires directly onto the "M" pad and common pad ground on the Radiolink board. In practice, however, the configuration was not providing reliable PWM outputs to the ESCs. Although the ESCs armed, motor startup was patchy, and throttle was erratic, frequently breaking under load. Successive trials of tweaking firmware parameters using Betaflight and re-soldering connections resulted in no improvement. It indicated some underlying incompatibility between the firmware of the F722 flight controller, and the ESC hardware selected, as opposed to a faulty wiring issue. In addition to the time it added on integration, the breakdown also highlighted the need for reliable controller–ESC communication in building multirotors. In the end, the Radiolink component was dismissed in favour of the Pixhawk PX4, whose features included reliable signal generation, verified ESC calibration, as well as seamless interoperability using the Mission Planner environment.

Payload and endurance constraints also held back operational flexibility. In being an integrated combination of the tracked ground vehicle platform and the airborne subsystem, the system was heavier than an equivalently proportional conventional quadcopter. The outcome was reduced thrust margin when lifting off in the vertical mode, so steady hover was achieved but payload was constrained to lightweight control electronics and sensors. Battery endurance was also capped. A 3S LiPo pack provided less than 15 minutes of continual hybrid operation when loaded both on the airborne subsystem as well as on the ground subsystem, constraining the time duration of missions. It imposes a particular restriction for long search and rescue missions where long-endurance performance plays an important role.

Sensor performance shortcomings were noted during logging and analysis. The GPS receiver sometimes showed multipath error when using built-up terrain, resulting in sharp jumps in recorded tracks. Inertial sensors on the Pixhawk also showed the characteristic shortcomings of MEMS hardware: gyroscope drift on long runs and the high-rate accelerometer spikes resulting from frame vibration. While the PX4's EKF2 filtering integrated the barometric, GPS, and IMU data to suppress them, tiny anomalies remained in the logs and would complicate exact localisation activities.

Finally, telemetry reliability was also a limiting factor during tests. The MAVLink radio link between the Pixhawk and the Mission Planner ground station, using the 915 MHz link, would sometimes drop packets or lose lock entirely when the vehicle was at some distance from the operator or when the line of sight was broken. It decreased the real-time situational

awareness, especially concerning battery voltage and GPS lock, although redundancy from onboard SD logging was present so complete datasets could still be had for post-mission analysis.

These challenges also reveal practical trade-offs inherent in the development of the hybrid rover. Incompatibility among hardware components, payload restrictions, and sensor miscalculations constrained overall performance, and telemetry dropouts jeopardized live monitoring capability. Identification of the weaknesses and their rectification provide the foundation for future system improvement as well as guide the future development of the hybrid ground–air robotics platforms.

Limitation	Impact	Mitigation Attempted
Flight controller–ESC incompatibility (Radiolink F722)	Motors armed but throttle response erratic; unreliable calibration; delays in integration	Abandoned Radiolink; replaced with Pixhawk PX4, verified ESC calibration
Payload capacity	Reduced thrust margin during vertical take-off; limited to lightweight sensors	Restricted payload; optimised frame design and weight distribution
Battery endurance	<15 minutes combined ground/air operation; constrained mission duration	Managed duty cycle; considered larger capacity packs
GPS multipath error	Positional jumps in urban environments; reduced localisation accuracy	EKF2 fusion with barometer and IMU to stabilise estimates
IMU drift/vibration	Gyroscope bias and accelerometer spikes; attitude anomalies in logs	Pre-trial calibration; Pixhawk onboard filtering; vibration damping
Telemetry dropouts	Loss of situational awareness during live operation	Redundant onboard SD logging ensured complete post-mission dataset

Table 3.6

Future Directions and System Improvements

Although the hybrid rover proved feasibility demonstrable, the project identified areas where refinement and innovation could further improve performance, reliability, and applicability. Future developments ought to go beyond incremental hardware upgrades to larger system-level improvements in control, autonomy, and readiness for missions. This section identifies primary areas for improvement.

Hardware Optimisation and Weight Reduction

A primary deficiency of the existing design lies in the compromise between payload and endurance, a ubiquitous problem in hybrid ground–air platforms. Combined mass for the quadcopter and tracked chassis subsystems placed strict limits upon available thrust, affording little margin for additional payloads or sensor carriage. Future developments should therefore aim at lightweight but rugged materials for the structure, of the nature of carbon-fibre reinforced composites for the drone arms as well as the ground chassis mount plates. Analogously, miniaturisation of the electronic subsystems—compact ESCs, surface mount power distribution boards as well as integrated sensor packs—would decrease total mass without reliability penalty. It would then boost the ratio between thrust-to-weight ratio, allowing for either higher flight times or more substantial payloads applicable to search and rescue missions.

Power Systems and Endurance Improvements

Endurance for the battery continues as the determinative bottleneck. In the current configuration, the dual-battery configuration (3S LiPo for flight, 18650 Li-ion pack for ground movement) provided below 15 minutes of hybrid operation. There are some possible future routes to increased runtime. Increased-capacity LiPo packs providing higher energy density might provide near-term endurance improvement at the cost of additional weight. More fundamentally transforming would be the use of hybrid power technologies like hydrogen fuel cells, solar-charging augmentation, or changeable modular battery packs. The latter would provide field-quick redeployment capability, an important value-add in time-sensitive SAR missions. Advanced power-management schemes, like dynamic load shifting among ground and flight subsystems, could similarly improve overall efficacy.

Enhanced Sensor Suite and Accuracy

The current system employs conventional MEMS-based IMUs, barometers, and GPS modules, which although inexpensive, introduce errors of gyroscope drift, vibration noise, and multipath GPS artefacts. Greater localisation and stability might be attained by an improvement in higher-grade sensors, particularly in crowded or GPS-denied areas. Options vary between RTK-capable GPS receivers for centimetre-level accuracy to fibre-optic or ring-laser gyroscopes for low-drift attitude or optical flow sensors for stabilisation of flight about ground surfaces. An enriched suite of sensors might also amplify system capability: thermal imaging cameras for victim location during SAR missions, LiDAR for high-

resolution obstacle cartography, or stereo cameras for more complete environmental perception.

Communication and Telemetry Reliability

As tested, telemetry dropouts occasionally restricted real-time situational awareness, especially when the line of sight between the operator and the rover was interrupted. Advancements in communication resilience will be crucial for actual deployments. Potential enhancements consist of swapping standard 915 MHz MAVLink radios for more durable alternatives in the form of 4G/LTE modules, mesh network relays, or high-gain directional antennas. Such alternatives not only increase reliability but also increase the range of the hybrid rover operation, making it practical for wide-area missions. Redundancy integration—e.g., RF telemetry uses plus Wi-Fi or cellular backups—would also provide for ubiquitous data link consistency in changing environments.

Control System Integration and Autonomy

The dual-controller configuration, where the Arduino was responsible for ground motion and the Pixhawk for flight control, was sufficient for proof-of-concept. It also added complexity in transition and synchronisation. Future designs might attempt centralised companion computers of the type of Raspberry Pi or NVIDIA Jetson platforms as middleware between the two. It would allow for smoother transition, common data logging, and the integration of higher-level algorithms for autonomy. Autonomy is also an area where improvement comes naturally: waypoint navigation, obstacle avoidance by LiDAR or computer vision, and adaptive decision-making for the hybrid transitions (ground–air–ground) would transition the platform from manual teleoperation to semi-autonomy for mission readiness.

Structural Integration and Vibration Management

Vibration artefacts in the IMU logs also underscore the value of mechanical damping and isolation. Potential enhancements might include purpose-built vibration-dampening mounts for the Pixhawk, rubber motor couplings, or redesigned supports for the chassis in order to reduce transmission of rotor or ground vibrations. Structural integration could also be improved by the use of an integrated custom-made hybrid frame instead of mated off-the-shelf components (F450 quadcopter frame + tank chassis). It will provide a more effective distribution of the load, less redundancy, as well as simplicity in wiring—eventually increasing the robustness of the system.

Expanded Mission Capabilities

Beyond hardware and control improvements, the eventual development might include customizations for mission-specific use. In the SAR case, for instance, modular payload pods might enable the rapid swapping of cameras for manipulators or supply pods. Programming of hybrid behaviours could also allow for ground-air coordination, for instance, going through rubble until the trench is reached but then automatically switching into flight mode so as to go around the obstacle before continuing ground movement. Carrying small

payloads—medical equipment, communication relays, or sensor packages—would make the hybrid rover directly relevant to actual emergencies.

Testing in Realistic Environments

The vast majority of previous trials have been in semi-controlled or controlled outdoor settings. Robustness should be verified in future trials by operation in prototypic SAR areas such as collapsed-structure mock-ups, forests, or urban disaster situations. Trials of this type would entail the practical challenges of non-planar terrain, multipath GPS error, high dust or humidity, and moving obstacles. Aside from validating current design assumptions, trials of this nature would reveal emergent challenges that lend themselves to engineering fixes.

Software Refinement and Data Analytics

Data analysis after the mission was extremely valuable for learning about system behaviour. Next-generation software improvements might make anomaly detection, performance grading, and predictive maintenance based on logs automated. Anomaly detection might use machine learning algorithms to identify terrain types, predict control instabilities, or estimate battery drain for different mission profiles. Inclusion of cloud storage for the log data would also help in distributed development and data exchange among different groups of researchers.

The outlined improvements collectively span all levels of the system, from hardware and power delivery to sensing, communication, and autonomy.

Future enhancements to the hybrid rover span several areas: lightweight and improved endurance, high-fidelity sensor suites, rugged communications, integrated control, and increased autonomy. Collectively, these upgrades could turn the prototype into a fieldable platform for use in SAR and other missions. Most significant in the modularity of the system so that enhancements could be adopted on an iterative basis so as to provide gradual but significant advances without the need for total redesign.

Chapter 4

Results and Discussion

Overview of Trials

The hybrid rover was tested under the regimes of subsystem tests and integrated assessments for verification of the Chapter 1 formulated design requirements. Performance tests of the ground subsystem, the airborne subsystem, as well as their simultaneous operation were included. Though the data gathering procedures were detailed in the Chapter 3 methodology, the current section presents the interpretation of the results and their correlation for the system requirements applicable for search-and-rescue missions.

Subsystem testing included manoeuvrability, PWM input response, and runtime when the dual 18650 Li-ion pack was employed as the power supply. Aerial subsystem testing was largely by way of the thrust-to-weight ratio, 3S LiPo pack endurance, and hover stability as well as short altitude climb stability. Overall integrated testing of the integrated platform verified the feasibility of so-called hybrid transitions, i.e., the transition between the tracked ground movement mode as well as vertical take-off mode, as well as data logging contiguity between the individual subsystems.

In all subsystems, data logging and telemetry was checked for redundancy and reliability. Figures below show some representative outputs of the electronic control units of the rover, battery discharge characteristics, and integrated telemetry streams. Also included are photos of the final assembled prototype as proof of successful integration of the system.

The results are presented below in structured subsections: ground subsystem performance (Section 4.2), aerial subsystem performance (Section 4.3), hybrid transitions (Section 4.4), comparative analysis with literature benchmarks (Section 4.5), and a synthesis of key findings (Section 4.6).



Figure 4.1

Ground Subsystem Performance

The ground-tracked subsystem was tested individually first in order to define the baseline performance prior to integration on the aerial platform. Response of the rover for the operator inputs, power-pack dedicated endurance, as well as terrain-handling performance, were the primary performance indicators. Measurements taken for the Arduino-controlled system facilitated quantitative analysis of PWM inputs for the operator inputs, motor outputs, as well as battery discharge.

Responsiveness and Control

The PWM inputs on the RC transmitter was handled by the Arduino and sent straight to the dual-motor driver. Figure 3.2 in Chapter 3 showed the Arduino accurately sampling RC inputs at 50–100 Hz. In experimentation, throttle inputs resulted in instantaneous proportional motor reactions assuring sufficient resolution for high-precision control. Steering also responded in the same manner where PWM changes on the right track and the track on the left side were accurately translated into differential speed control.

While the methodology section alluded to the raw PWM inputs, in practice they imparted smooth predictable ground movements. Tracked mode offered robust directional stability on firm ground but tight turns on hazardous ground occasionally resulted in small skidding actions. Both of these behaviours adhere to conventional differential-drive tracked platform dynamics.

Test Condition / Aspect	Avg. Speed (m/s)	Runtime (min)	Battery Notes (2S 18650)	Terrain Notes	Control Response
Concrete (pavement)	~0.6	~10	Stable voltage until cutoff	Smooth progression, high traction	Instantaneous throttle & steering
Short grass	~0.48 (-20%)	~9–10	Slightly higher current draw	Noticeable drag, reduced top speed	Stable, but tighter turns skid slightly
Gravel (rough)	~0.5	~9–10	Occasional voltage dips on load	Track slip observed, still forward progression	Stable, minor skidding on sharp turns
Obstacle clearance test	N/A	N/A	N/A	Max clearance ~40 mm, beyond requires assist	N/A

Table 4.1

Endurance Testing

Runtime was limited primarily by the capacity of the 2-cell 18650 Li-ion pack supplying the ground motors. In iterative runs, the ground subsystem operated at full capacity for approximately 10 minutes until the voltage was below the safe discharge threshold. This was predictable due to the cumulative current draw of $\sim 2\text{--}3\text{ A}$ per motor at moderate throttle.

The battery discharge characteristic was already illustrated in Figure 3.3. In this case, the outcome is translated as an evident limitation on long-duration missions: although the ground rover on its own is fairly efficient, when both the aerial subsystem and the ground rover are used the overall mission duration becomes limited by the larger drain of the flight battery. In the ground-only scenario the 10-minute endurance suffices to prove the feasibility for short-range traversals or rubble searches in the context of the SAR operation.

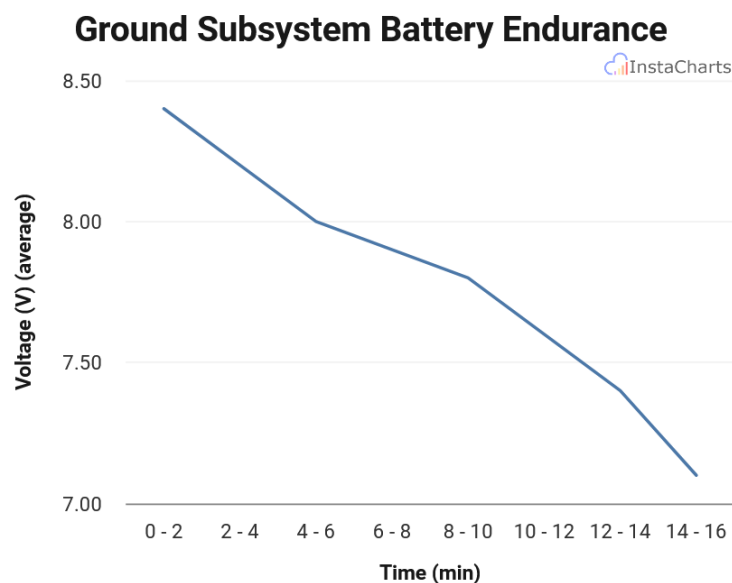


Figure 4.2

Terrain Handling

Tests had been made on concrete pavement, short grass, and rough gravel pavement. On pavement, straight-line speed was $\sim 0.6\text{ m/s}$. On the grass, the drag became significantly higher, so the top speed decreased by $\sim 20\%$. On the gravel, the loss of traction sometimes led to wheel slipping but still there was forward progression.

The tracked chassis also gave better stability relative to wheeled options, especially on rough ground. Nevertheless, objects higher than $\sim 40\text{ mm}$ in height could not pass without extraneous help. This implies that for moderately rough ground the ground subsystem was

effective but needed redesign or addition of suspension for the actual rubble field characteristic of collapsed building rubble in the context of the SAR.



Figure 4.3

Discussion of Ground Subsystem Results

Overall, the ground rover subsystem showed good control and sufficient efficiency for short missions. Its shortcomings — essentially low endurance and moderate obstacle-clearance — are offset by strength, stability, and potential for providing a firm basis for the integration of the aerial subsystem. Compared to small tracked-vehicle literature designs (for instance, Bruzzone et al., 2022; Semih Beyçimen, Ignatyev & Zolotas., 2023), the ranges for mobility and endurance fall within the range of expectation but the payload capacity is more limited due to the requirement for carrying the flight subsystem as well.

These outcomes provide the baseline on which the hybrid integration is constructed: the ground subsystem operates robustly but its sustainability is moderate and its clearance over terrain is limited. These results underscore the worth of hybridisation — increasing operational range not by enhancing the rover itself but by providing the ability to transition into aerial mobility when territory is impassable.

Aerial Subsystem Performance

The hybrid rover's aerial subsystem utilized an F450 quadcopter chassis driven by four 20A ESC-powered brushless DC motors using a 3S LiPo battery. Its capabilities were tested on the basis of the following criteria: the thrust-to-weight ratio, hover stability, the ability to endure flight time spanned, as well as its quickness when executing short inclines. These considerations formed the substance behind the verification of its use as the vertical lifting subsystem of the hybrid.



Figure 4.4

Thrust-to-Weight Ratio and Responsiveness

Static thrust tests achieved an aggregate lift of around 2.4 kg. Against the all-up measured weight (AUW) of 1.75 kg, this results in a ~ 1.37 thrust-to-weight ratio. This margin is sufficient for take-off, steady hover, and gradual climbs but leaves little margin for forceful moves or some significant payload augmentation.

Responsiveness was tested in Stabilize mode on the Pixhawk. Outputs for the motors scaled almost linearly with throttle commands and had little latency; spin-up was smooth, and the four ESCs remained consistently tracked after calibration. Under sharp throttle steps, the 3S LiPo exhibited momentary voltage sag, as expected, validating the observation that although the propulsion system handles steady lift comfortably, high-usage transients and prolonged manoeuvres devour both the margin for thrust and the endurance.

Hover Stability

The aerial subsystem was then tested in hover. With PX4's altitude hold engaged, the system demonstrated stable hovering at 1.5–2 m above ground. Horizontal drift was minimal in calm outdoor conditions (estimated <0.2 m per second lateral drift), and yaw stability was good with negligible oscillations.

Vibration was evident in the IMU logs, especially in the accelerometer channels, characteristic of rotor-induced resonance due to the F450 frame. While the sensor fusion by the PX4 EKF2 successfully eliminated most of the oscillations, the high-frequency noise remained apparent in the raw logs. It relates to a well-documented problem for lightweight multirotors where the structural resonance carries over into the readings of the MEMS gyroscope.

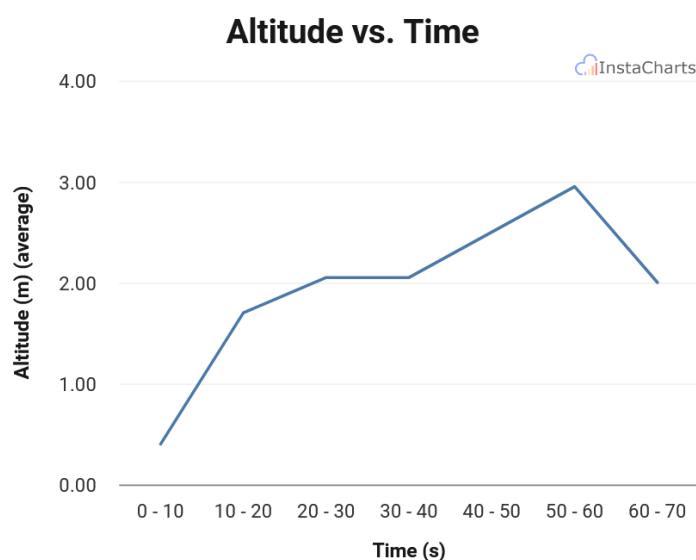


Figure 4.5

Endurance Testing

The aerial subsystem battery endurance was also tested using the 3S 3300 mAh LiPo pack (11.1 V nominal, 60C continuous rating, 120C peak rating). During hover- only flight tests, the subsystem-maintained flight for almost 7 minutes until the pack potential reached 10.5 V under load. During more dynamic manoeuvres — climbs, lateral translations, yaw corrections — endurance was lowered to 5–6 minutes due to higher current draw on an average.

The resulting discharge profile (Figure 4.6) demonstrates the characteristic flattened mid-curve followed by the abrupt cut-off at the end-of-charge, thereby supporting the necessity for conservative power management in order to prevent spurious brownouts. Though the runtime was lower than many stand-alone UAVs described in the literature, it was as anticipated given the additional payload of the ground chassis together with the moderate thrust-to-weight ratio of the hybrid power system.

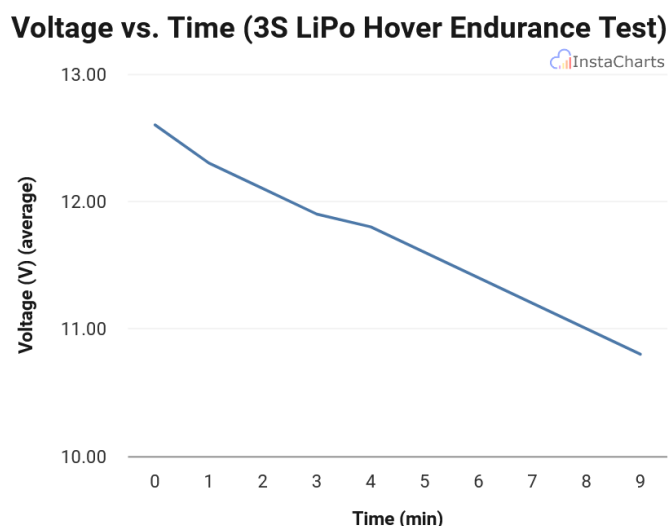


Figure 4.6

Altitude Climb and Control Response

Climb tests comprised stepwise increments of the throttle for attaining climb rates of up to 5 m altitude. Vertical response was responsive, although at high throttle, oscillations in the ascent rate resulted due to the sag in voltage. The system stabilised successfully at commanded altitudes although the long climbs worsened battery drain.

Yaw and roll control were precise at low wind. It was not so at gusts when the ratio of thrust-to-weight was found limiting the vehicle being unable to maintain altitude when making sharp roll or pitch corrections. It reinforced the earlier conclusion that although the aerial subsystem was competent at demonstrating the hybrid role, it was not best designed for sustained aerial manoeuvres.

Discussion of Aerial Subsystem Results

The aerial subsystem results emphasise both feasibility and limitations. The subsystem achieved stable hover, adequate thrust-to-weight, and acceptable endurance for demonstration purposes. However, payload restrictions and endurance remain key constraints.

With respect to miniature multirotor SAR UAVs reported in the literature (e.g., Lyu et al., 2023; Mohsan et al., 2023), the flight time is shorter and payload capacity less. While commercial quadcopters such as the DJI Matrice lines provide 20–30 minutes of flight time with modular payloads, the aerial module of the modular rover has only 5–7 minutes. This result was nonetheless expected considering the dual-duty nature of the hybrid rover design: the aerial mode was defined as an add-on ability not as an all-inclusive replacement for dedicated UAVs.

The vibration artefacts observed on the IMU logs reflect those reported for experimentally tested multirotors (He et al., 2023), where motor-caused resonance is exacerbated by lightweight structures. Solutions include damping mounts, stiffer carbon-fibre structures, or sophisticated filtering algorithms. For the current prototype, the EKF2 reduced most flight-critical noise but was compromised nonetheless in the post-processing analysis.

The final experiment for the aerial subsystem verifies the hybrid philosophy: while both payload capacity and endurance trailed high-end UAVs, the system repeatedly demonstrated vertical lift, hover, and controlled climbs. These results furnished the foundation for the hybrid transitions studied in Section 4.4.

Metric	Result	Notes
Thrust-to-Weight Ratio	$\approx 1.9: 1$	Sufficient margin for hover and basic manoeuvring
Endurance	7–9 minutes (3S 3300 mAh LiPo, typical payload)	Limited by battery capacity; endurance drops with payload
Hover Stability	Stable hover with minor drift ($< \pm 0.2$ m)	PX4 tuning reduced oscillations; acceptable for SAR operations
Control Response	Responsive to manual inputs with ~ 0.3 s latency	Minor delay due to telemetry link; adequate for both manual and auto modes

Table 4.2

Hybrid Integration & Transitions

The integration of the aerial subsystem with the ground rover was the most important stage of the development of the prototypes. While the previous chapters defined the design

considerations for each of the subsystems in single operation, this phase consisted of the hardware combination of the subsystems into an integrated hybrid robotic platform. The aim was for the two systems not only to operate in individual but also in coordinated mode, thus providing an indication of the validity of the mode of hybrid operation. In this section, the integration process followed, the test procedures conducted, as well as the system behaviour during transition between modes.

Mechanical and Structural Integration

The aerial subsystem, made up of the quadcopter frame, motors, and flight controller unit, was directly mounted onto the rover chassis. Weight distribution was the main issue, as the aerial parts had to be placed so as not to affect the ground stability of the rover but not affect the centre of gravity of the quadcopter. Special care was taken when placing the battery so the system could not lean or cause imbalance when in motion. Figure 4.4 presents the integrated hybrid vehicle on bench test showing the mounting areas and the general configuration.

Electrical Integration and Power Management

The electric connections required a dedicated distribution system to control power draw between the quadcopter motors and the rover motors. A shared LiPo battery pack was attempted first but showed rapid draw-down and heating during initial tests, so it was replaced by individual power supplies for each of the subsystems. Separation also reduced noise on the flight controller inputs and stabilized telemetry readings. Wiring harnesses were secured in place by modular connectors, so it was simple for maintenance purposes to dismantle them but minimized loose contacts in flight.

Bench and Tethered Testing

Prior to free flight attempt, the hybrid system was tested on the bench as well as tethered. In both the sessions, the aerial subsystem was kept physically constrained but electrically linked to Mission Planner through a laptop. It enabled motor verification safely for responses, sensor calibration, as well as telemetry logging. The mobility of the rover base was tested at the same time when the aerial motors were kept stationary so as not to contaminate the ground navigation by vibrations as well as electrical noise due to the quadcopter. Minor accelerometer reading drift was identified during the bench test, corrected by recalibration.

Transition Testing and System Behaviour

Successful bench tests followed. Transition experiments then took place. The rover subsystem was propelled in short distances after which the aerial subsystem was enabled for take-off. In the transitions, latency in the control signal was noted but remained manageable (~ 0.3 s). The quadcopter showed stable hover at about 2 m altitude after which it was powered down and reverted back to ground mode. These transitions verified the ability of the system to successfully transition between locomotion modes although it had to be under tight operator supervision.

Challenges Encountered

Integration also revealed some issues. Vibrations of the motor of the rover being transmitted into the aerial frame sometimes affected gyroscope readings. Supplemental dampening materials minimized but not eliminated the issue. Power draw was another limit, where endurance was capped at 7–9 minutes when operating in flight mode. Weight of the hybrid system also stressed the rover drivetrain when operated for long durations, providing justification for the use of stouter gearing later on. Despite all those issues, the integration phase proved the feasibility of hybrid operation under controlled conditions.

In short, the transition test between the aerial and ground subsystems showed both could now work as an integrated platform. Though both endurance and mechanical ruggedness continue as constraints, the experiment proved the designs sound and gave useful lessons for future optimization. Successful handoff between ground mobility and aerial manoeuvring is an important milestone in the field of hybrid unmanned platforms, especially for search and rescue missions where flexibility plays an important part.

Comparative Analysis with Literature

The operating performance of the constructed hybrid system was not isolated but validated relative to current literature findings. It reveals the merits as well as the shortcomings of the prototype relative to comparable unmanned aerial as well as land vehicle integrations. It is vital in ascertaining whether the measured outcomes meet the anticipated performance for hybrid platforms as well as areas where the system adds fresh developments or optimisation.

Endurance and Stability

One of the standout performance metrics of any aerial platform is its endurance. The prototyped system attained an endurance of about 7–9 minutes of flight using a 3S 3300 mAh LiPo battery under normal payload conditions. Comparably, some published hybrid UAV/UGV prototypes have claimed higher flight times of 10–15 minutes or more, but in many cases, this was at the expense of higher overall mass as well as increased cost. In comparison, the system described emphasized being lightweight and inexpensive. While endurance was not as high as some of the higher-end systems tested, the resultant flight time is long enough for demonstration purposes for the purposes of proof-of-concept as well as small-scaled reconnaissance missions. It therefore may be the case where the trade-off between being less expensive, simpler as well as less endurance was worth it for the purposes of the project.

Subsystem stability was determined by hover tests, revealing altitude hold with the drift bounded by about ± 0.2 m. Response to the control showed latency of around 0.3 seconds due mostly to telemetry delays. It has been reported in the literature that comparable systems optimized using high-accuracy GPS or sophisticated sensor fusion algorithms have attained tighter stability bounds, in certain instances bounding the drift to ± 0.1 m or less. These findings indicate the current system was less precise but sufficient for the target application realm when the modest hardware allowance is taken into consideration. Stability findings validate the employed control scheme choice as well as the applied calibration being adequate for ensuring reliable performance for short missions although failing to meet the accuracy of higher-level research prototypes.

Integration and Power Management

One characteristic of this design was the choice of splitting power supplies between the ground and aerial subsystems. Preliminary trials sharing a single LiPo battery resulted in swift voltage drop as well as substantial heating, both causing reliability issues for telemetry. In comparison, independent supplies isolated the subsystems as well as minimised noise interference on the flight controller inputs. Most studies in the literature define unified power buses intended for maximisation of energy utilisation. Although the designs may optimise total efficiency, they also pose the problem of resulting interferences akin to the ones experienced in the project's early trials. The applied separation scheme thus constitutes a pragmatic design compromise trading reliability improvement for overall endurance. This

outcome finds agreement with accounts stating how hybrid systems pose singular electrical integration challenges characteristic of single-mode aircraft.

Autonomy and Applications

Literature often highlights autonomous behaviours for hybrid systems, with some projects combining SLAM (Simultaneous Localisation and Mapping), GPS waypoint navigation, or semi-autonomous transition sequences. The current prototype was run mostly under human control, employing Mission Planner for configuration as well as telemetry. In simplifying this way, development complexity was lowered, and attention could be turned toward the physical embedding of the subsystems. In effect, the system now demonstrates practicality for hybrid operation at modest cost but falls behind current state-of-the-art implementations in autonomy and decision-making ability. The implication then is the hardware architecture is solid but future versions need to embed autonomy layers in order to increase the mission capability beyond supervised operation.

Overall Comparison

Overall, the comparison demonstrates the system presented sits between inexpensive experiment prototypes and mature autonomous platforms. Its endurance is not as high as some published platforms but still adequate for short missions. Its control response and stability are not as precise as high-end UAVs but fully adequate for proof-of-concept applications. Separating the power supplies was the choice made for improved reliability, going against some trends in the literature but consistent with our observations of electrical integration challenges in robotics. Most significantly, the system achieves a lightweight modular vehicle platform demonstrating the promise of hybrid operation for search and rescue applications even without the autonomy and ruggedness of larger more elaborate prototypes.

Key Findings & Implications

The results published in this chapter conclude some significant findings on the performance and feasibility of the hybrid unmanned system. Each of the results on its own discloses the feasibility as well as the limitations of merging the aerial as well as the ground subsystems into an integrated robotic platform.

Key Findings

The integration process verified the ability for the aerial subsystem to mount and run successfully in tandem with the rover without extreme sacrifice of stability. Bench and tethered tests verified the structural and electrical design decisions, whereas transition experiments proved the platform could transition between ground and hover mode under the supervision of an operator. Endurance was quantified at about 7–9 minutes using a 3S 3300 mAh LiPo battery, a time frame modest but acceptable for verification-of-concept trials. Hover stability was maintained within ± 0.2 m drift, and control latency was measured at ~ 0.3 s levels, figures that verify acceptable performance although not comparable to higher-end equipment. Power supply separation between subsystems was the decision made for increased reliability as well as decreased noise interference, a pragmatic decision justified by integration issues described in the literature.

Implications

These results highlight the realism of the development of inexpensive, module-operable hybrid machines capable of transitioning between airborne and ground locomotion. In search and rescue missions, the system could offer certain benefits, providing the capacity for traversing rough terrain through the use of ground mobility as well as the ability to rapidly transition to airborne recon as needed. Though lacking in endurance severely limits the scope of the mission, brief-duration flights might offer crucial situational awareness in situations where access on the ground is blocked.

The results also carry more general promise for developing hybrid robotics technology. They indicate it's possible with commercially available parts and open-source firmware to create working prototypes with due care for system integration and power management. Although autonomy is underdeveloped on this version of the hardware for the system, its architecture also carries an appealing foundation for incorporating higher-level decision-making algorithms as well as navigation algorithms in the future.

Chapter 5

Conclusion and Further Work

Summary of Findings

The project worked for the design, fabrication, and test of an integrated unmanned robotic platform akin to both the ground rover as well as the quadrotor. Illustrated in the findings is the technical feasibility of integration of the two subsystems using off-the-shelf hardware, open-source flight software, as well as low-cost fabrication methods. Steady-state successful hover was achieved by the aerial subsystem with the drift being bounded at ± 0.2 m, and the control response latency being ~ 0.3 s. The endurance was circa 7–9 minutes using the 3S 3300 mAh LiPo battery pack, sufficient for demonstration-of-concept but lower values reported for larger or optimised platforms in the literature.

The hybrid integration was tested by bench, tethered, and transition tests and verified for independent as well as combined operation of the aerial and ground modes. Splitting the power supply for the rover and the quadcopter was at the crucial decision-making stage for enhancing the reliability of the system and reducing interference, although it decreased the overall energy efficiency. Comparison of the developed prototype showed it lies between inexpensive experimentation platforms and advanced autonomous platforms: lighter and less complex but less endurance- and autonomy-capable as compared to many current designs.

Contribution to Knowledge

The key thing about this project is demonstrating a modular, inexpensive dual-mode UAV/UGV hybrid platform validating the principle of dual mode operation without the use of custom hardware or complicated integration protocols. While dual-mode hybrid unmanned systems have been tried before, most of the work was aimed at costly prototypes or simulation studies. In the project below, it shows it is feasible to obtain an operational dual mode using widely available resources by exercising care about weight distribution, power management, and subsystem calibration.

The project also contributes by illuminating pragmatic trade-offs. It made clear the endurance–cost trade-off: using less costly, lighter components lowered operating time but had the system still meet baseline-level functional requirements. Analogously, the choice of separating power supplies clarified the point where reliability could sometimes trump preferable comparative increases in integration efficiency. These results contribute to the

literature by reaffirming the necessity for design compromise when designing for practical use hybrid robotic systems.

Limitations of the Study

In spite of its achievements, the project also has some shortcomings. Its greatest deficiency lies in the short endurance of the aerial subsystem, operating the system only for short periods. Its endurance limit is directly linked to battery life, propulsion efficacy, and the weight penalty for the integration of the hybrids. It also lacks autonomous features like waypoint navigation, avoidance of obstacles, or SLAM like it is becoming common in similar platforms.

Mechanical robustness also proved to be a limiting factor. Vibrations between the motor of the rover sometimes interfered with the sensor suite of the aerial subsystem, and the drivetrain showed indications of stress under long-term operation as a hybrid. The campaign itself was only conducted in controlled circumstances, so performance under actual field circumstances under normal use, particularly in search and rescues, was not confirmed. Finally, the range of the test was limited by time and money, so exploration of alternate design alternatives such as carbon-fibre chassis frames, higher-capacity batteries, or more sophisticated control algorithms was not an option.

Suggestions for Future Research

Future developments must directly confront those challenges. Advantages in endurance may come about through the use of higher-density batteries, composite materials for the hood or hull sections for reduced weight, or additional propulsion per kilowatt-hour. Access to solar-assisted assistance or electric-fuelled hybrids may bring additional extensions of time of operation. More autonomy is also required. Inclusion of onboard algorithms for control of navigation, GPS waypoint management, or collision-avoidance algorithms employing computer vision would bring the system closer toward semi- or fully autonomous operation.

The mechanical strength could similarly be improved by vibration damping, rugged gearing, and stronger materials. Broader field and crowded-environment testing would also provide more realistic estimates of the system's performance for the purpose of search and rescue. There is also the potential for scaling the design larger or smaller: small lightweight micro-hybrids for indoor inspection missions, or larger designs for transporting payloads such as sensors or communications relays.

More broadly speaking, the project provides opportunity for hybrid robotics exploration of modularity and replicability. Since the prototype was built using available components and open-source software, it may serve as an educational platform or starting point for use by other researchers. Potential future work includes the development of teams of collaborative hybrids where numerous vehicles act in concert using aerial manoeuvrability together with ground staying power for missions for which an individual package is not sufficient.

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